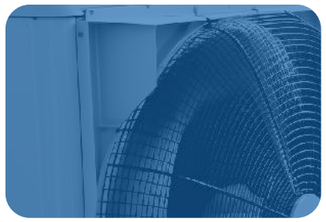
## CONTACT US


MINGGUANG HECHENG ELECTRICAL CO.,LTD.
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Hvacstar






# HECHENG ELECTRICAL

20 years experience for HVAC/R parts
More Professional
More Reliable
More Helpful

Since 2004

# Our Vision



To be the most professional manufacturer of motor control and protection products in the electrical industry

Our Mission:

Committed to serving customers more professionally, more reliably and more effectively

Our Values:

harmonious,integrity,pioneer,innovation

Mingguang Hecheng Electrical Co.,Ltd.has been working on developing high quality Definite Purpose Contactor, Disconnet Switch,Relays and other control parts for HVAC system

Since estalished in 2004.We have experience in supplying for many famous companies and brands directly or indirectly.







1.We have an industrial park of 70000m²

2.We have a strong team of engineers and experienced staff.

3.Our markets cover North America, South America, Middle East, Africa, Europe and Asia.

4.Our products have passed UL&CSA&SEMKO&CE&CB&CCC and other certifications









## 24/7 Emergency Callout

20 years OEM production experience ensure quality products with quick delivery time.

## Fully Qualified & Insured

We adopt ISO 9001:2015 System, We use PLM+ERP System for R&D and Production Control ensure quality.

## Certification

Before market, Whole products approved by UL,CSA,SEMKO, CCC,CE and necessary test.



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# SA Series Contactors





## Product overview

SA Series DP Contactor is mainly used to control motor load for AC 50Hz/60Hz ,rated insulation voltage: 690V ,rated work current:90A Air Conditioner compressor, also applicable for other household and similar purposes device.

## Features&benefits

- ◇ Small size, compact structure, same installation dimensions, easy for replace.
- ◇ High heat-resistant coil, suitable for both 50Hz/60Hz & wide voltage range.
- ◇ Contact with excellent Arc resistance capacity.
- ◇ Resist Arc effectively with patented Arc extinguishing technology.

## Product certification

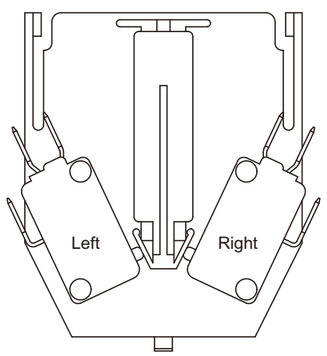
| | | | | | |
|---|---|---|---|---|---|
| c UL us | UL | UL 60947-4-1 | CE | CE | IEC 60947-4-1 |
| c SP us | CSA | C22.2 No.60947-4-1 | A2L | A2L | CSA C22.2 No.60335-2-40<br>UL 60335-2-40 |
| S | Semko | IEC 60947-4-1 | RoHS | RoHS | RoHS Compliant |
| CCC | CCC | GB/T 14048.4 | Reach | REACH | REACH Compliant |

## Code explanation

| | SA | Y | 2X | B | 30 | AA | * |
|---|---|---|---|---|---|---|---|
| Basic Model Designation: SA | | | | | | | |
| Pack Code: | | Normal factory bulk packaging for OEM or volume delivery skid quantities, not for individual sale or distribution;<br>- : Normal factory bulk packaging<br>Y: Individual box packaging | | | | | |
| Number of Pole: | | | 1N: 1 pole normally open double mark not with shunt<br>1X: 1 pole normally open double mark with shunt<br>2X: 2 poles normally open double make<br>3X: 3 poles normally open double make<br>4X: 4 poles normally open double make<br>2E: 2Poles normally close | | | | |
| Coil Voltage: | | | | B: 24 VAC 50/60Hz G: 120 VAC 50/60Hz U: 208/240 VAC 50/60Hz<br>W: 277 VAC 50/60Hz E: 380 VAC 50/60Hz F: 480 VAC 50/60Hz H:36VDC only for 1 Pole | | | |
| Contact Rating (Full Load Amps-FLA): | | | | | 20: 20A, 25: 25A, 30: 30A, 35: 35A, 40: 40A, 50: 50A 60: 60A; 75: 75A 90: 90A | | |
| Termination Code: | | | | | | AA: Screw mode and quick connect terminal for 1P&2P series<br>CC: Frame mode and quick connect terminal for 1P&2P series<br>CY: Screw mode and quick connect terminal for 3P (FLA 20-40A) &4P series<br>CG: Frame mode and quick connect terminal for 3P (FLA 20-40A) &4P series<br>CJ: Frame mode and quick connect terminal for 3P (FLA 50-90A) series<br>BB: Screw mode and quick connect terminal except optional SEMS plate | |
| Special Order | | | | | | | |

## Code explanation

### Auxiliary contact(Micro Switch)




SA | 2C | 300V | Q | L

Product Series
SA: apply SA series

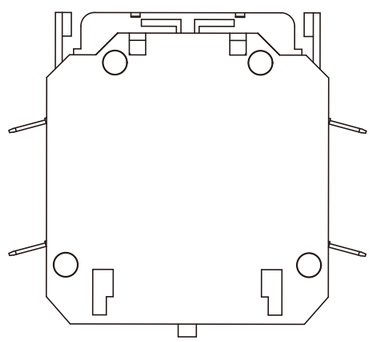
Contact form and Qty
1C：1 SPDT
2C：2 SPDT

Maximum Voltage level
300V：250V AC/DC

Terminal Code
Q：Quick connect terminal

Location of Micro Switch
Nil：Left+Right
L：Left
R：Right

### Auxiliary contact



SA | 1Z | 600V | Q

Product Series
SA: apply SA series

Contact form
1X：1NO　2X：2NO
1Y：1NC　2Y：2NC
1Z：1NO+1NC

Maximum Voltage Level
600V：600VAC

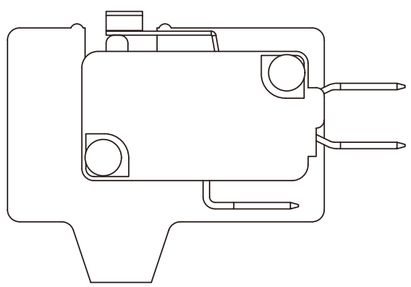
Terminal Code
Q：Quick connect terminal
S：Screw terminal
C：QC+Screw (Only 1 NO or 1NC)



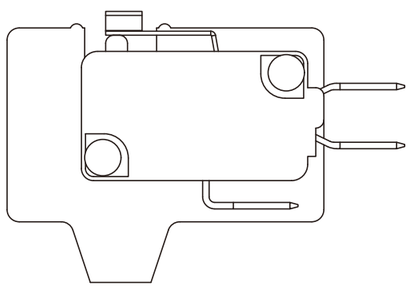
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SA Series Contactors

### Auxiliary contact(Micro Switch)

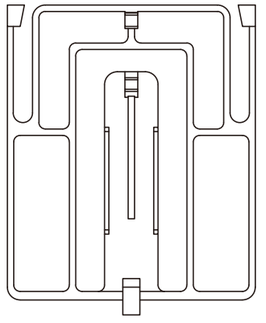


SA1C300VQ6



SA1C300VQ9

### Mechanical Interlock



SAMI-40

## Ambient condition

| | | |
|---|---|---|
| Ambient temperature (℃) | Operating | -40～+65 |
| | Storage | -40～+70 |
| Relative humidit | 90%～95%RH（Below 40℃） | |
| Pollution degree | 3 | |
| Enclosure rating | IP00 | |
| Max Working Altitude（m） | 2000 | |
| Installation | M5 Screw installation，Horizontal or vertical installation of the base | |

## Technical Parameter

CONTACT RATING - Main Contact

| Full Load Amps FLA | RES | LRA | | | |
|---|---|---|---|---|---|
| | | 240V | 277V | 480V | 600V |
| A | A | A | A | A | A |
| 1 POLE 20-40FLA(Include 1+N) | | | | | |
| 20 | 30 | 120 | 120 | 100 | 80 |
| 25 | 35 | 150 | 150 | 125 | 100 |
| 30 | 40 | 180 | 180 | 150 | 120 |
| 40 | 50 | 240 | 240 | 200 | 160 |
| 2 POLES 20-90FLA | | | | | |
| 20 | 30 | 120 | 120 | 100 | 80 |
| 25 | 35 | 150 | 150 | 125 | 100 |
| 30 | 40 | 180 | 180 | 150 | 120 |
| 40 | 50 | 240 | 240 | 200 | 160 |
| 50 | 65 | 300 | 240 | 240 | 200 |
| 60 | 75 | 360 | 240 | 300 | 240 |
| 75 | 90 | 450 | 300 | 300 | - |
| 90 | 120 | 540 | 360 | 360 | - |



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SA Series Contactors

CONTACT RATING - Main Contact

| Full Load Amps FLA | RES | LRA | | | | Max Power | | |
|---|---|---|---|---|---|---|---|---|
| | | 240V | 277V | 480V | 600V | Voltage | 1 Hp | 3 Hp |
| A | A | A | A | A | A | V | hp | hp |
| 3&4 POLES 20-40FLA | | | | | | | | |
| 20 | 30 | 120 | 120 | 100 | 80 | 120V | 1.5 | – |
| | | | | | | 240V | 3 | 7.5 |
| | | | | | | 480V | - | 7.5 |
| | | | | | | 600V | - | 7.5 |
| 25 | 35 | 150 | 150 | 125 | 100 | 120V | 2 | - |
| | | | | | | 240V | 5 | 7.5 |
| | | | | | | 480V | - | 10 |
| | | | | | | 600V | - | 15 |
| 30 | 40 | 180 | 180 | 150 | 120 | 120V | 3 | - |
| | | | | | | 240V | 5 | 10 |
| | | | | | | 480V | - | 15 |
| | | | | | | 600V | - | 20 |
| 40 | 50 | 240 | 240 | 200 | 160 | 120V | 3 | - |
| | | | | | | 240V | 7.5 | 10 |
| | | | | | | 480V | - | 20 |
| | | | | | | 600V | - | 25 |
| 3 POLES 50-60FLA | | | | | | | | |
| 50 | 65 | 300 | 240 | 240 | 200 | 120V | 3 | – |
| | | | | | | 240V | 7.5 | 15 |
| | | | | | | 480V | - | 25 |
| | | | | | | 600V | - | 25 |
| 60 | 75 | 360 | 240 | 300 | 240 | 120V | 5 | - |
| | | | | | | 240V | 15 | 20 |
| | | | | | | 480V | - | 30 |
| | | | | | | 600V | - | 30 |
| 3 POLES 75-90FLA | | | | | | | | |
| 75 | 90 | 450 | 300 | 300 | - | 120V | 5 | – |
| | | | | | | 240V | 15 | 25 |
| | | | | | | 480V | - | 40 |
| | | | | | | 600V | - | 50 |
| 90 | 120 | 540 | 360 | 360 | - | 120V | 7.5 | - |
| | | | | | | 240V | 20 | 30 |
| | | | | | | 480V | - | 50 |
| | | | | | | 600V | - | 60 |

## CONTACT IEC RATING - Main Contact

| Product specifications | AC-1 | AC-3 | | AC-8a | AC-8b | Ith(A) |
|---|---|---|---|---|---|---|
| | 240VAC | 240VAC | 277VAC | 240VAC | 240VAC | |
| 1 Pole 20 FLA | 30 | 20 | 20 | 20 | 7 | 30 |
| 1 Pole 25 FLA | 35 | 25 | 25 | 25 | 8 | 35 |
| 1 Pole 30 FLA | 40 | 30 | 30 | 30 | 10 | 40 |
| 1 Pole 40 FLA | 50 | 40 | 40 | 40 | 15 | 50 |
| 2 Poles 20 FLA | 30 | 20 | 20 | 20 | 7 | 30 |
| 2 Poles 25 FLA | 35 | 25 | 25 | 25 | 8 | 35 |
| 2 Poles 30 FLA | 40 | 30 | 30 | 30 | 10 | 40 |
| 2 Poles 40 FLA | 50 | 40 | 40 | 40 | 15 | 50 |

| Product specifications | AC-1 | AC-3 | | AC-8a | AC-8b | Ith(A) |
|---|---|---|---|---|---|---|
| | 415VAC | 415VAC | 690VAC | 415VAC | 415VAC | |
| 3 Poles 20 FLA | 30 | 20 | 20 | 20 | 7 | 30 |
| 3 Poles 25 FLA | 35 | 25 | 25 | 25 | 8 | 35 |
| 3 Poles 30 FLA | 40 | 30 | 30 | 30 | 10 | 40 |
| 3 Poles 40 FLA | 50 | 40 | 40 | 40 | 15 | 50 |
| 3 Poles 50 FLA | 65 | 50 | 32 | 50 | 18 | 65 |
| 3 Poles 60 FLA | 75 | 60 | 32 | 60 | 22 | 75 |
| 3 Poles 75 FLA | 90 | 75 | 39 | 75 | 27 | 90 |
| 3 Poles 90 FLA | 120 | 90 | 47 | 90 | 34 | 120 |
| 4 Poles 20 FLA | 30 | 20 | 20 | 20 | 7 | 30 |
| 4 Poles 25 FLA | 35 | 25 | 25 | 25 | 8 | 35 |
| 4 Poles 30 FLA | 40 | 30 | 30 | 30 | 10 | 40 |
| 4 Poles 40 FLA | 50 | 40 | 40 | 40 | 15 | 50 |







## CONTACT RATING - Auxiliary Contact

| Contact Form: | Rating Voltage (VAC) | 120 | 240 | 480 | 600 |
|---|---|---|---|---|---|
| 1 NO 2NO<br>1 NC 2NC<br>1 NO+1 NC | Rating breaking current (A) | 3 | 1.5 | 0.75 | 0.6 |
| | Maximal making current (A) | 30 | 15 | 7.5 | 6 |
| | Ith (A) | 10 | 10 | 10 | 10 |

## CONTACT RATING - Micro Switch

| Contact Form: | |
|---|---|
| 1 SPDT | 16A,125 or 250VAC<br>1/3hp,125VAC<br>10A,30VDC |

## Coil Rating(+25°C)

| Product specifications | MaxInrush (V·A) | | Max Seal (V•A) | | Voltage (VAC) | | |
|---|---|---|---|---|---|---|---|
| | 60Hz | 50Hz | 60Hz | 50Hz | Operate | Dropout | Max Voltage |
| SA 1 Pole 20-40FLA | 20 | 25 | 6.5 | 7.5 | ≤0.8Us | ≥0.2Us | 1.1Us |
| SA 2 Poles 20-40FLA | 45 | 35 | 7.5 | 9 | | | |
| SA 3 Poles 20-40FLA | 50 | 60 | 7 | 8 | | | |
| SA 4 Poles 20-40FLA | 55 | 65 | 7 | 8 | | | |
| SA 3 Poles 50-60FLA | 120 | 140 | 23 | 28 | | | |
| SA 3 Poles 75-90FLA | 180 | 200 | 30 | 35 | | | |

Remark:

1.Coil Us=24VAC 50/60Hz or 120VAC 50/60Hz or 208/240VAC 50/60Hz or 277VAC 50/60Hz, 380VAC 50/60Hz or 480VAC 50/60Hz.

2.Starting and holding capacity are both measured at the rated voltage of the coil Us.

3.Pick-up & drop-off voltage are tested under the condition of the base being perpendicular to the horizontal (vertical installation).

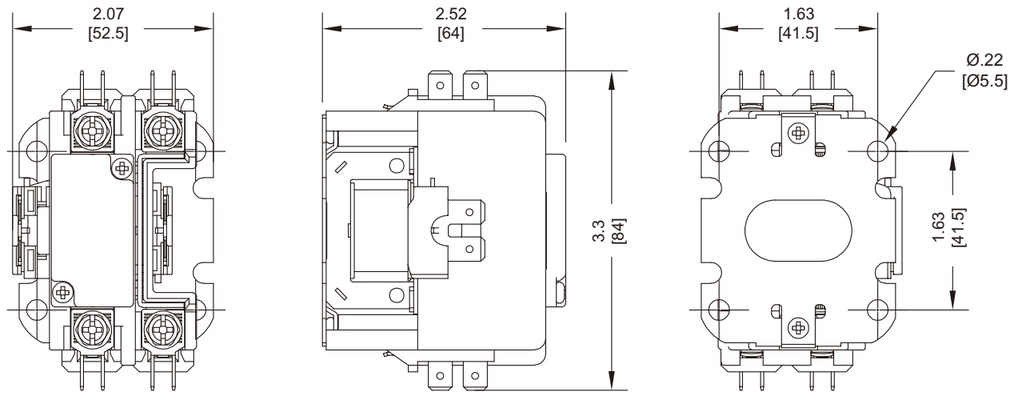
## Technical parameter description

Above data is based on the testing under standard condition with manufacturer's sample , actual use may vary due to differences in environment, on/off load, on/off frequency, and equipment.

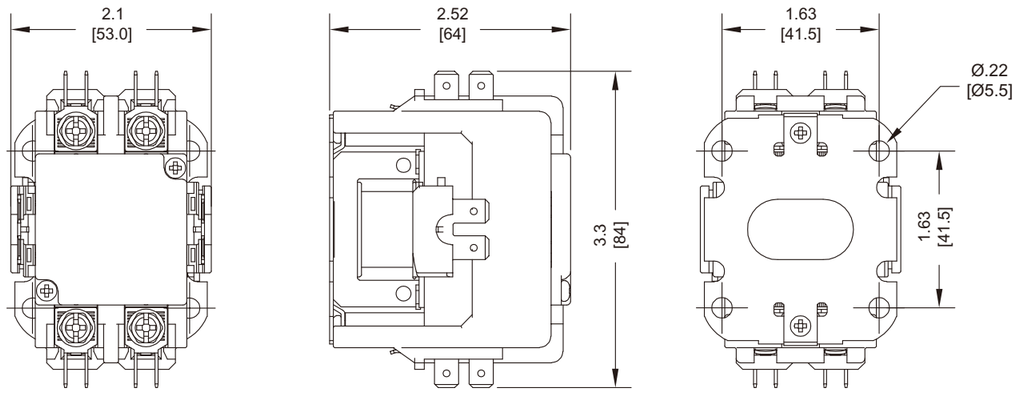
## Dimension

inch[mm]

### SA-1P 20-40A




### SA-2P 20-40A



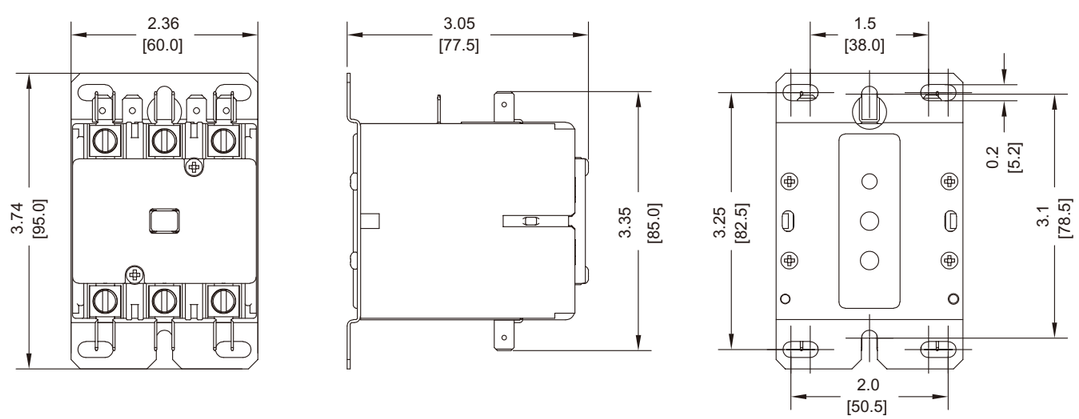




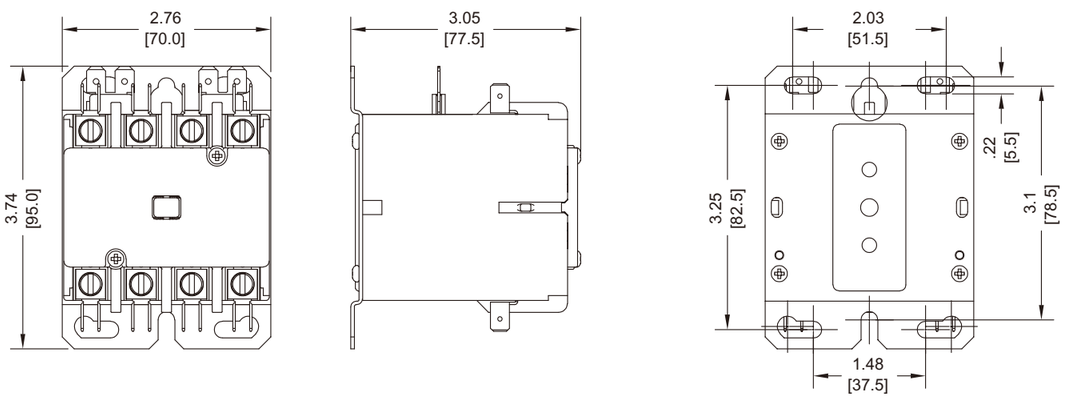



inch[mm]

### SA-3P 20-40A

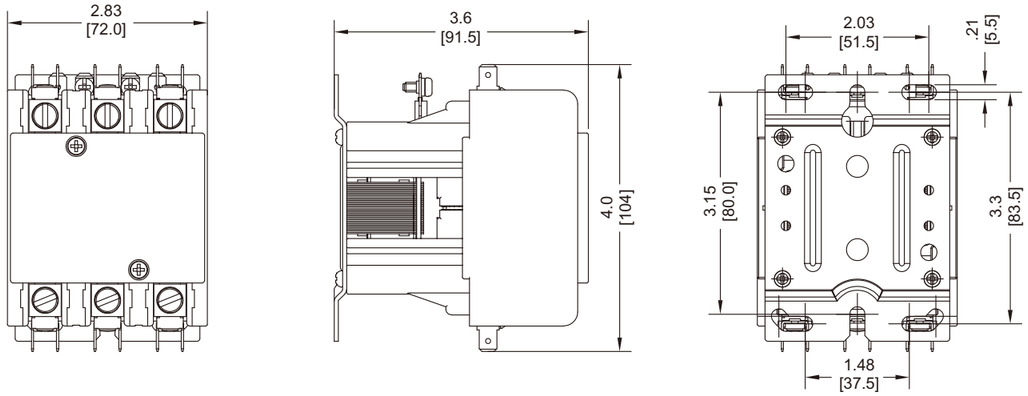



### SA-4P 20-40A

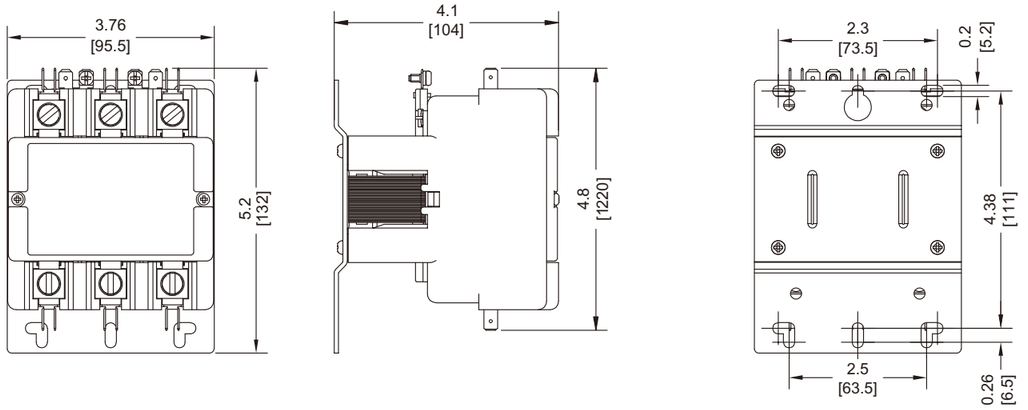


inch[mm]

## SA-3P 50-60A




## SA-3P 75-90A



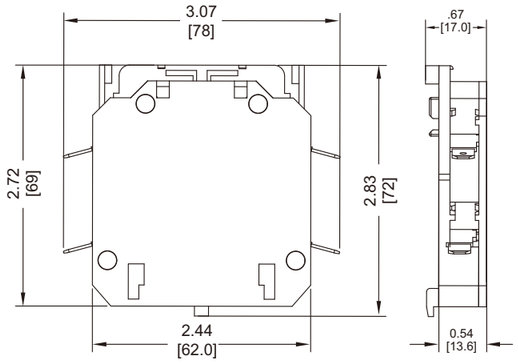







inch[mm]

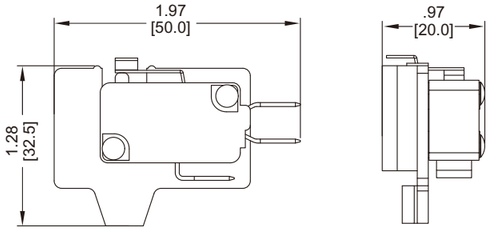
## Auxiliary Contact-Apply to 3&4 Poles 20-40FLA



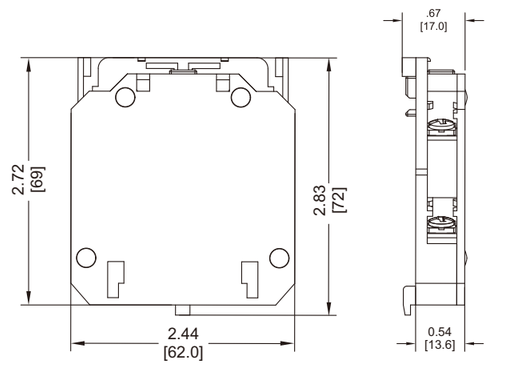

## Auxiliary Contact(Micro Switch) -Apply to 3&4 Poles 20-40FLA



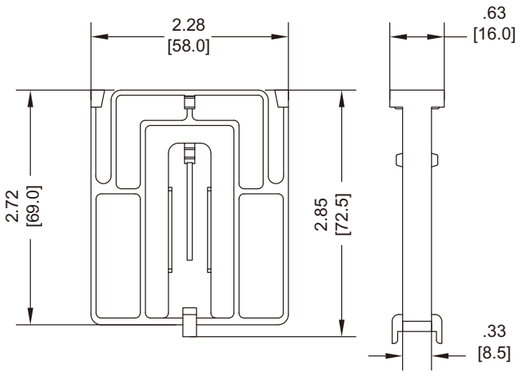

## Auxiliary Contact （Micro Switch） -Apply to 3 Poles 50-60FLA

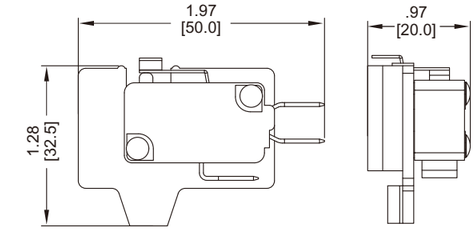



## Mechanical Interlock -Apply to 3&4 Poles 20-40FLA

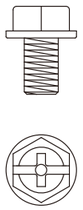



## Auxiliary Contact （Micro Switch） -Apply to 3 Poles 75-90FLA

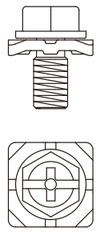


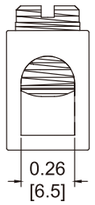
## Terminals



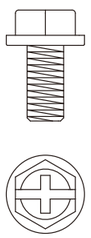
20-40FLA
#10-32 Combination Screw



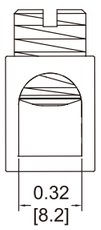
20-40FLA
#10-32 SEMS Screw



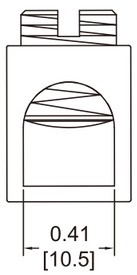

20-40FLA——Lug
#14-4 AWG Cu/Al



50-60FLA
Combination Screw




50-60FLA——Lug
#14-2 AWG Cu/Al




75-90FLA——Lug
#14-1 AWG Cu only

Remark:
1.The tolerance of wiring connector without specified dimensions shall be in accordance with ±0.2mm.
2.Main circuit screw locking torque for rated current 20-40FLA products: SCREWS 22 IN-LBS / LUGS 40 IN-LBS Main circuit screw locking torque for rated current 50-60FLA products: 50 IN-LBS
Main circuit screw locking torque for rated current 75-90FLA products: 50 IN-LBS
Auxiliary module screw locking torque: 10 IN-LBS
Coil screw locking torque for rated current 20-90FLA products: 10 IN-LBS
3.The quick connect socket terminal must comply with IEC60760 standard and ensure reliable connection between terminal and insert ,if insert wiring is used.

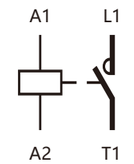




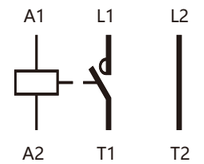


## Diagram

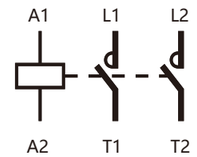
### Contactor



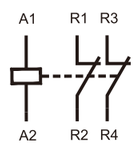

1 Pole 20-40 FLA



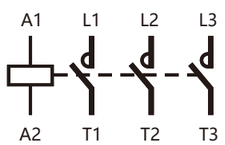

1+N Pole 20-40 FLA



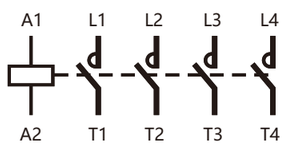

2 Poles 20-90 FLA




2 Poles 20-40 FLA




3 Poles 20-90 FLA

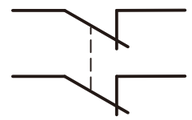



4 Poles 20-40 FLA

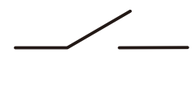
### Auxiliary Contact



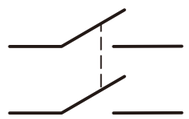
1NC



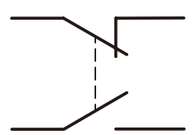
2NC



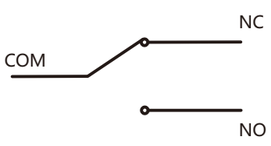
1NO



2NO



1NO+1NC




SPDT

# HCDP Series Contactors





Hvacstar

## Product overview

HCDP Series DP Contactor is mainly used to control motor load for AC 50Hz/60Hz ,rated insulation voltage： 690V ,rated work current:90A Air Conditioner compressor, also applicable for other household and similar purposes device.

## Features&benefits

◇ Small size, compact structure, same installation dimensions, easy for replace.

◇ High heat-resistant coil, suitable for both 50Hz/60Hz & wide voltage range.

◇ Contact with excellent Arc resistance capacity.

◇ Resist Arc effectively with patented Arc extinguishing technology.

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| A2L | A2L | CSA C22.2 No.60335-2-40<br>UL 60335-2-40 |
| RoHS | RoHS | RoHS Compliant |
| Reach | REACH | REACH Compliant |

## Code explanation

| | HCDP | Y | 2X | B | 30 | AA | * |
|---|---|---|---|---|---|---|---|

Basic Model Designation: HCDP

Pack Code:
Normal factory bulk packaging for OEM or volume delivery skid quantities, not for individual sale or distribution;
- : Normal factory bulk packaging
Y: Individual box packaging

Number of Pole:
1N: 1 pole normally open double mark not with shunt
1X: 1 pole normally open double mark with shunt
2X: 2 poles normally open double make
3X: 3 poles normally open double make
4X: 4 poles normally open double make
3E: 2 poles normally open, 1 poles normally close
4E: 2 poles normally open, 2 poles normally close

Coil Voltage:
B: 24 VAC 50/60Hz G: 120 VAC 50/60Hz U: 208/240 VAC 50/60Hz
W: 277 VAC 50/60Hz E: 380 VAC 50/60Hz F: 480 VAC 50/60Hz

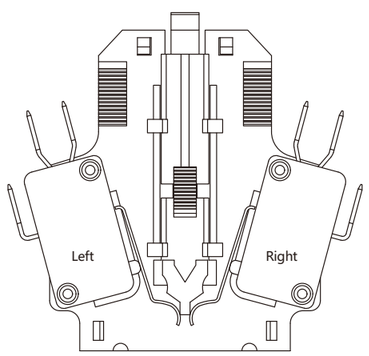
Contact Rating (Full Load Amps-FLA):
20: 20A, 25: 25A, 30: 30A, 35: 35A, 40: 40A,
50: 50A 60: 60A;75: 75A 90: 90A

Termination Code:
AA: Screw mode and quick connect terminal for 1P&2P series
CC: Frame mode and quick connect terminal for 1P&2P series
CY: Screw mode and quick connect terminal for 3P (FLA 20-40A) &4P series
CG: Frame mode and quick connect terminal for 3P (FLA 20-40A) &4P series
CJ: Frame mode and quick connect terminal for 3P (FLA 50-90A) series
BB: Screw mode and quick connect terminal except optional SEMS plate

Special Order

## Code explanation

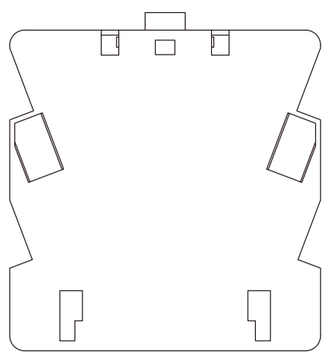
### Auxiliary contact(Micro Switch)




HCDP | 2C | 300V | Q | L

- Product Series
  HCDP: apply HCDP series
- Contact form and Qty
  1C：1 SPDT
  2C：2 SPDT
- Maximum Voltage level
  300V：250V AC/DC
- Terminal Code
  Q：Quick connect terminal
- Location of Micro Switch
  Nil：Left+Right
  L：Left
  R：Right

### Auxiliary contact



HCDP | 1Z | 600V | Q

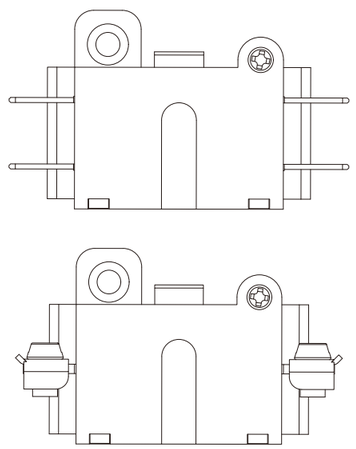
- Product Series
  HCDP: apply HCDP series
- Contact form
  1X：1NO　　2X:2NO
  1Y：1NC　　2Y:2NC
  1Z：1NO+1NC
- Maximum Voltage Level
  600V：600VAC
- Terminal Code
  Q：Quick connect terminal
  S：Screw terminal
  C：QC+Screw







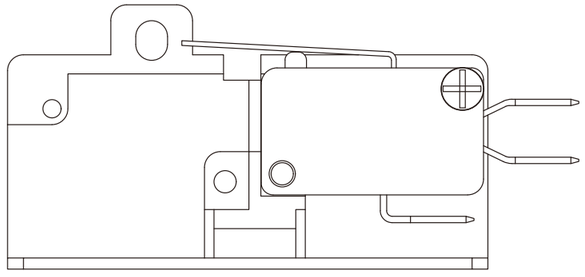
### Auxiliary contact



HCDP | 1Z | 600V | Q | 9

- Product Series
  HCDP: apply HCDP series
- Contact form
  1X：1NO
  1Y：1NC
  1Z：1NO+1NC
- Maximum Voltage Level
  600V：600VAC
- Terminal Code
  Q：Quick connect terminal
  S：Screw terminal (1NO or 1NC)
- Apply to 75-90FLA

### Auxiliary contact(Micro Switch)



HCDP1C300VQ9

## Ambient condition

| | | |
|---|---|---|
| Ambient temperature (℃) | Operating | -40～+65 |
| | Storage | -40～+70 |
| Relative humidit | 90%～95%RH（Below 40℃） | |
| Pollution degree | 3 | |
| Enclosure rating | IP00 | |
| Max Working Altitude（m） | 2000 | |
| Installation | M5 Screw installation，Horizontal or vertical installation of the base | |

## Technical Parameter

CONTACT RATING - Main Contact

| Full Load Amps FLA | RES | LRA | | | |
|---|---|---|---|---|---|
| | | 240V | 277V | 480V | 600V |
| A | A | A | A | A | A |
| 1 POLE 20-40FLA(Include 1+N) | | | | | |
| 20 | 30 | 120 | 120 | 100 | 100 |
| 25 | 35 | 150 | 150 | 125 | 100 |
| 30 | 40 | 180 | 180 | 150 | 120 |
| 35 | 40 | 210 | 210 | 175 | 160 |
| 40 | 50 | 240 | 240 | 200 | 160 |
| 2 POLES 20-90FLA | | | | | |
| 20 | 35 | 120 | 120 | 100 | 100 |
| 25 | 35 | 150 | 150 | 125 | 100 |
| 30 | 40 | 180 | 180 | 150 | 120 |
| 35 | 40 | 210 | 210 | 175 | 160 |
| 40 | 50 | 240 | 240 | 200 | 160 |
| 50 | 65 | 300 | 300 | 250 | 200 |
| 60 | 75 | 360 | 360 | 250 | 240 |
| 75 | 90 | 450 | 450 | 375 | 300 |
| 90 | 120 | 540 | 540 | 375 | 360 |







CONTACT RATING - Main Contact

| Full Load Amps FLA | RES | LRA | | | | Max Power | | |
|---|---|---|---|---|---|---|---|---|
| | | 240V | 277V | 480V | 600V | Voltage | 1 Hp | 3 Hp |
| A | A | A | A | A | A | V | hp | hp |
| 3&4 POLES 20-40FLA | | | | | | | | |
| 20 | 35 | 120 | 120 | 100 | 100 | 120V | 1.5 | – |
| | | | | | | 240V | 3 | 7.5 |
| | | | | | | 480V | - | 7.5 |
| | | | | | | 600V | - | 7.5 |
| 25 | 35 | 150 | 150 | 125 | 100 | 120V | 2 | - |
| | | | | | | 240V | 5 | 7.5 |
| | | | | | | 480V | - | 10 |
| | | | | | | 600V | - | 15 |
| 30 | 40 | 180 | 180 | 150 | 120 | 120V | 3 | - |
| | | | | | | 240V | 5 | 10 |
| | | | | | | 480V | - | 15 |
| | | | | | | 600V | - | 20 |
| 35 | 40 | 210 | 210 | 175 | 160 | 120V | 3 | - |
| | | | | | | 240V | 5 | 10 |
| | | | | | | 480V | - | 15 |
| | | | | | | 600V | - | 20 |
| 40 | 50 | 240 | 240 | 200 | 160 | 120V | 3 | - |
| | | | | | | 240V | 7.5 | 10 |
| | | | | | | 480V | - | 20 |
| | | | | | | 600V | - | 25 |
| 3 POLES 50-60FLA | | | | | | | | |
| 50 | 65 | 300 | 300 | 250 | 200 | 120V | 3 | - |
| | | | | | | 240V | 7.5 | 15 |
| | | | | | | 480V | - | 25 |
| | | | | | | 600V | - | 30 |
| 60 | 75 | 360 | 360 | 250 | 240 | 120V | 5 | - |
| | | | | | | 240V | 15 | 20 |
| | | | | | | 480V | - | 30 |
| | | | | | | 600V | - | 40 |
| 3 POLES 75-90FLA | | | | | | | | |
| 75 | 90 | 450 | 450 | 375 | 300 | 120V | 5 | - |
| | | | | | | 240V | 15 | 25 |
| | | | | | | 480V | - | 40 |
| | | | | | | 600V | - | 50 |
| 90 | 120 | 540 | 540 | 375 | 360 | 120V | 7.5 | - |
| | | | | | | 240V | 20 | 30 |
| | | | | | | 480V | - | 50 |
| | | | | | | 600V | - | 60 |

## CONTACT RATING - Auxiliary Contact

| Contact Form: | Rating Voltage (VAC) | 120 | 240 | 480 | 600 |
|---|---|---|---|---|---|
| 1 NO 2 NO<br>1 NC 2 NC<br>1 NO+1 NC | Rating breaking current (A) | 3 | 1.5 | 0.75 | 0.6 |
| | Maximal making current (A) | 30 | 15 | 7.5 | 6 |
| | Ith (A) | 10 | 10 | 10 | 10 |

## CONTACT RATING - Micro Switch

| Contact Form: | 10A,125 or 250VAC<br>0.6A,125VDC<br>0.3A,250VDC |
|---|---|
| 1 SPDT | |

## Coil Rating(+25℃)

| Product specifications | Max Inrush (V·A) | | Max Seal (V•A) | | Voltage (VAC) | | |
|---|---|---|---|---|---|---|---|
| | 60Hz | 50Hz | 60Hz | 50Hz | Operate | Dropout | Max Voltage |
| HCDP 1 pole 20-40FLA | 20 | 25 | 6.5 | 7.5 | ≤0.8Us | ≥0.2Us | 1.1Us |
| HCDP 2 poles 20-40FLA | 20 | 25 | 6.5 | 7.5 | | | |
| HCDP 3 poles 20-40FLA | 50 | 60 | 7 | 8 | | | |
| HCDP 4 poles 20-40FLA | 50 | 60 | 7 | 8 | | | |
| HCDP 3 poles 50-60FLA | 75 | 85 | 8.5 | 10 | | | |
| HCDP 3 poles 75-90FLA | 200 | 250 | 40 | 45 | | | |

Remark:

1.Coil Us=24VAC 50/60Hz or 120VAC 50/60Hz or 208/240VAC 50/60Hz or 277VAC 50/60Hz, 380VAC 50/60Hz or 480VAC 50/60Hz.

2.Starting and holding capacity are both measured at the rated voltage of the coil Us.

3.Pick-up & drop-off voltage are tested under the condition of the base being perpendicular to the horizontal (vertical installation).

Above data is based on the testing under standard condition with manufacturer's sample , actual use may vary due to differences in environment, on/off load, on/off frequency, and equipment.



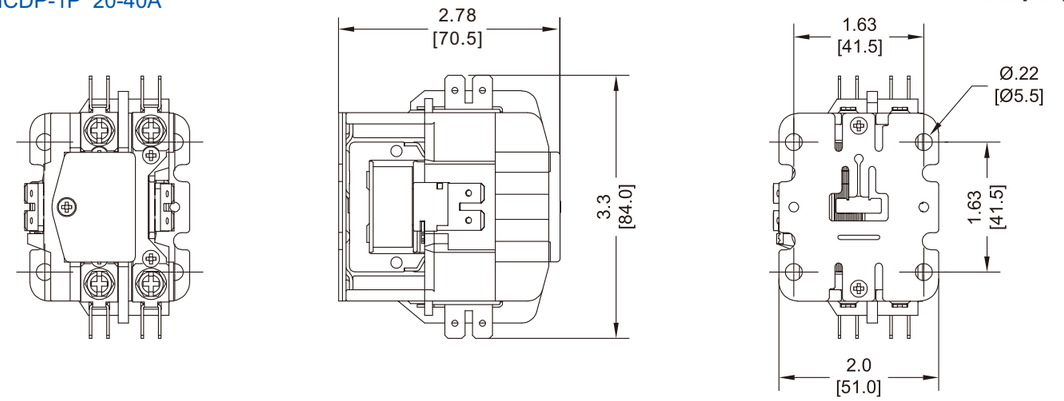




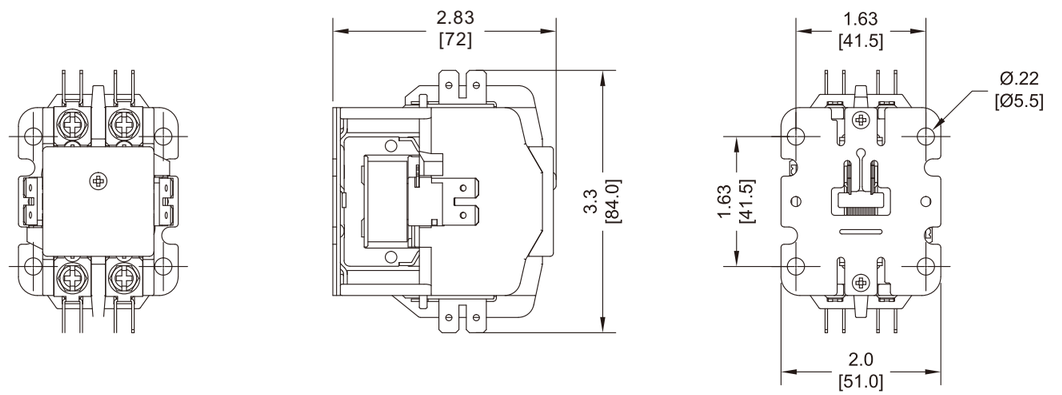
## Dimension

inch[mm]

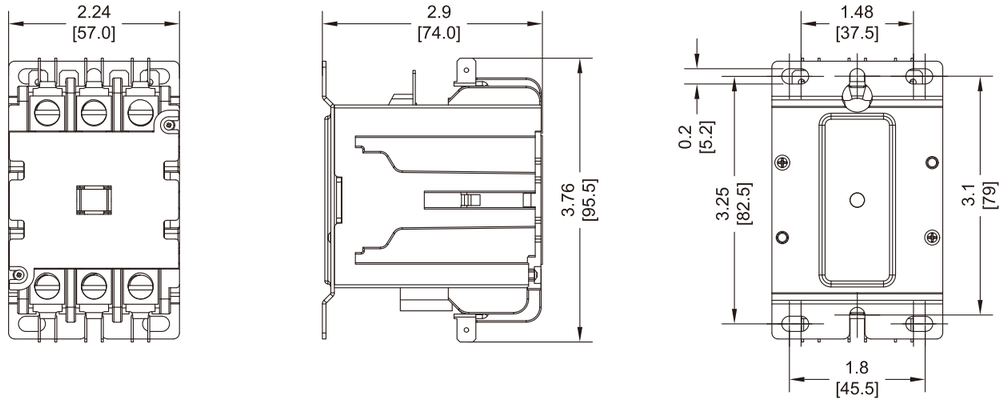
HCDP-1P 20-40A




HCDP-2P 20-40A

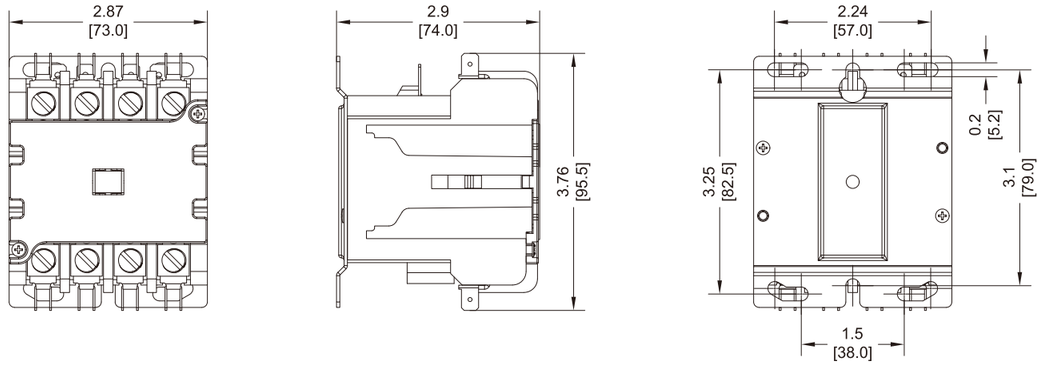



HCDP-3P 20-40A

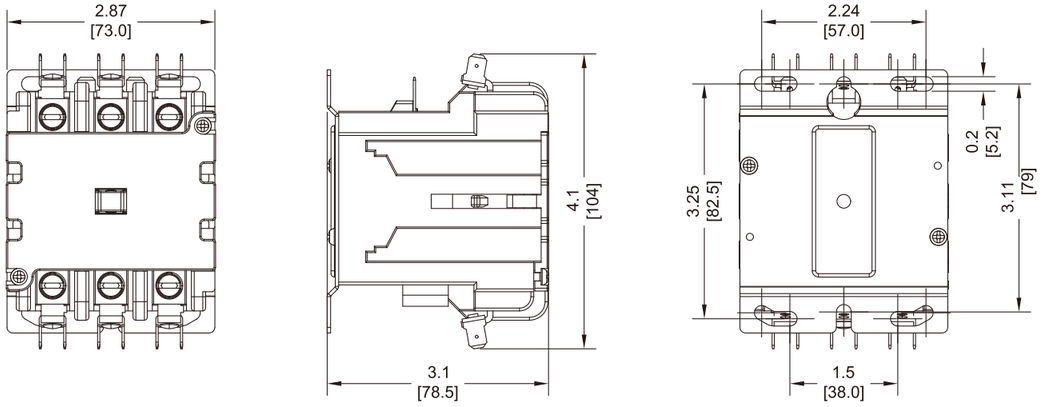


inch[mm]

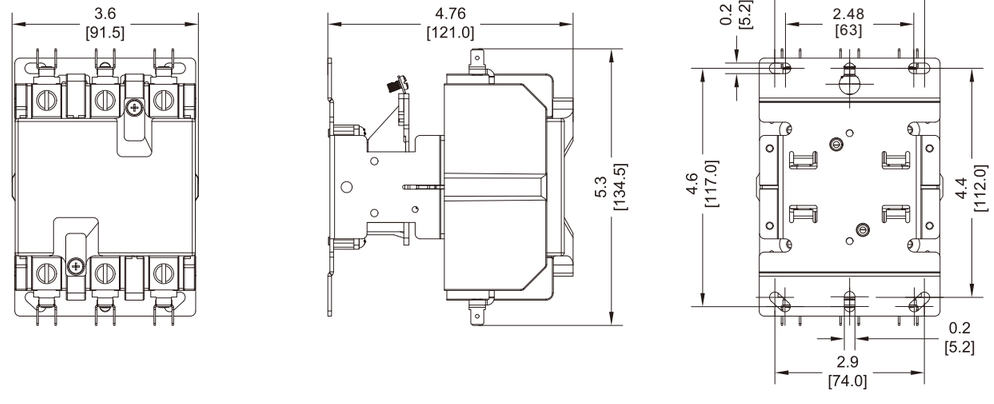
## HCDP-4P 20-40A




## HCDP-3P 50-60A




## HCDP-3P 75-90A



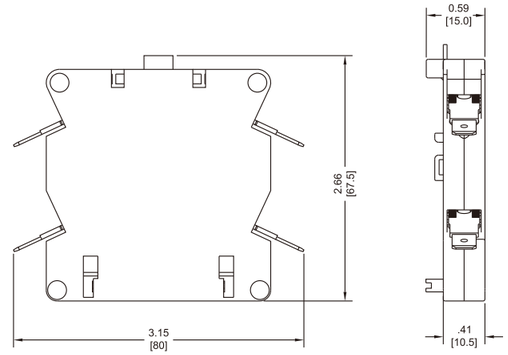




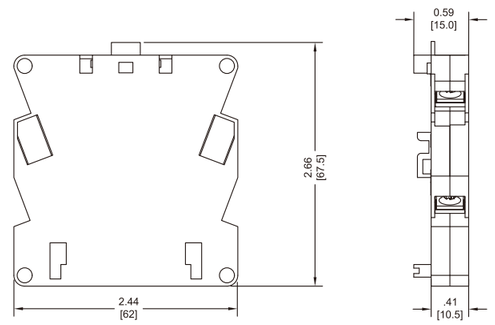



inch[mm]

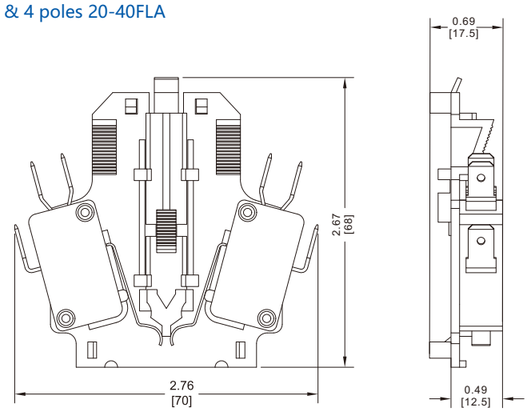
## Auxiliary Contact-Apply to 3 Poles 20-60FLA & 4 poles 20-40FLA

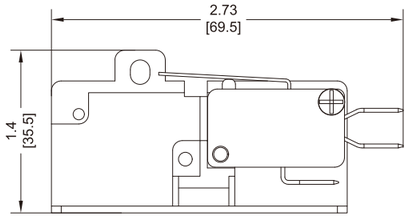



## Auxiliary Contact（Micro Switch）-Apply to 3 Poles 75-90FLA

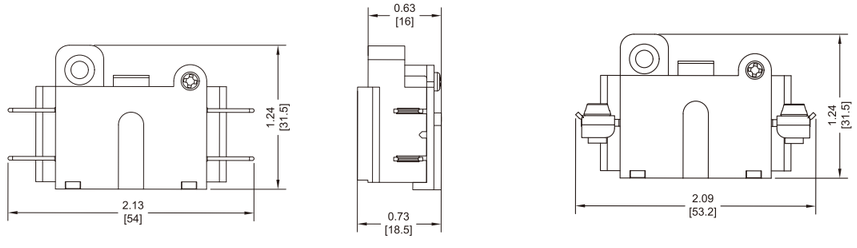



## Auxiliary Contact(Micro Switch)-Apply to 3 Poles 20-60FLA & 4 poles 20-40FLA



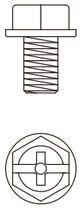




## Auxiliary Contact-Apply to 3 Poles 75-90FLA

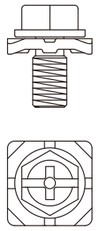


## Terminals

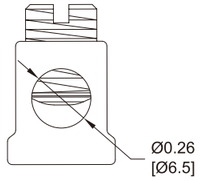
inch[mm]



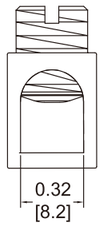
20-40FLA
#10-32 Combination Screw



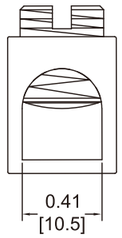
20-40FLA
#10-32 SEMS Screw




20-40FLA——Lug
#14-4 AWG Cu/Al




3 Poles 20-60FLA & 4 Poles 20-40FLA——Lug
#14-2 AWG Cu/Al




75-90FLA——Lug
#14-1 AWG Cu only

Remark:
1.The tolerance of wiring connector without specified dimensions shall be in accordance with ±0.2mm.
2.Main circuit screw locking torque for rated current 20-40FLA 1&2 Poles products: SCREWS 22 IN-LBS / LUGS 40 IN-LBS
Main circuit screw locking torque for rated current 20-40FLA 3&4 Poles and50-60FLA products: 50 IN-LBS
Main circuit screw locking torque for rated current 75-90FLA products: 50 IN-LBS
Auxiliary module screw locking torque: 10 IN-LBS
Coil screw locking torque for rated current 20-90FLA products: 10 IN-LBS
3.The quick connect socket terminal must comply with IEC60760 standard and ensure reliable connection between terminal and insert ,if insert wiring is used.

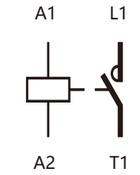




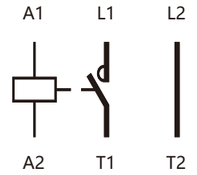


## Diagram

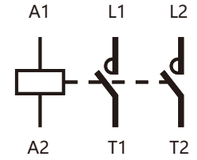
### Contactor



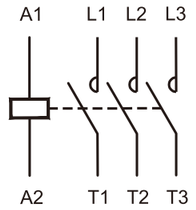

1 Pole 20-40 FLA



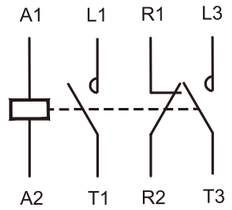

1+N Pole 20-40 FLA



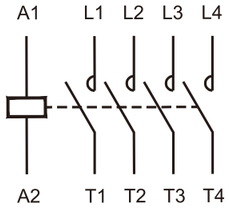

2 Poles 20-90 FLA

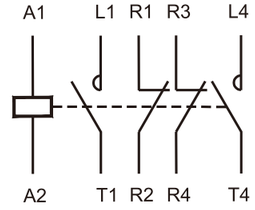



3 Poles 20-90 FLA




3 Poles 20-40 FLA



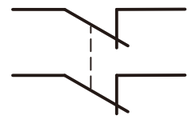




4 Poles 20-40 FLA

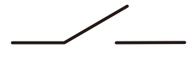
### Auxiliary Contact



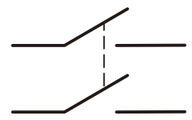
1NC



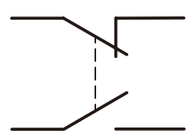
2NC



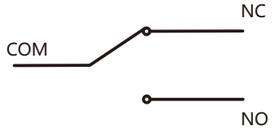
1NO



2NO



1NO+1NC




SPDT






Hvacstar



## Product overview

HCH8 contactor can be used for remote control:Lighting, heating, ventilation, rolling curtains,public hot water Mechanical ventilation system, etc Low inductive micro-inductive load environment

## Features&benefits

- ◇ Low power consumption
- ◇ Low noise
- ◇ Super long Electrical life

## Product certification

| | | |
|---|---|---|
| CCC | **CCC** | GB/T 14048.4 GB/T 17885 |
| CE | **CE** | IEC 60947-4-1 |
| RoHS | **RoHS** | RoHS Compliant |

## Code explanation

| HCH8 | 8 | 16 | 10 | AC230 | Z |
|---|---|---|---|---|---|

HCH8 — Modularization AC Contactor

8 — Design Serial Number

16 — Rated Current
16A/20A/25A/32A/40A/63A

10 — Contactor Type
1P 10:1NO/01:1NC
2P 20:2NO/11:1NO+1NC/02:2NC
3P 30:3NO/03:3NC
4P 40:4NO/31:3NO+1NC/22:2NO+2NC/04:4NC

AC230 — Control Voltage
AC24V/AC48V/AC110V/AC230V/DC12V/DC24V
Nil：AC Control Circuit

Z：DC Control Circuit

## Ambient condition

| | | |
|---|---|---|
| Ambient temperature (℃) | Operating | -5-+60 |
| | Storage | -40～+70 |
| Relative humidit | 90%～95%RH（Below 55℃） | |
| Pollution degree | 3 | |
| Enclosure rating | IP20 | |
| Max Working Altitude（m） | 2000 | |
| Installation | 35mm DIN | |

## Characteristic Parameters

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Rated Operational Current | AC-7a | A | 16 | 20 | 25 | 32 | 40 | 63 |
| | AC-7b | A | 6 | 7 | 9 | 12 | 15 | 20 |
| Conventional Free Air Thermal Current | | A | 25 | 25 | 25 | 63 | 63 | 63 |
| Rated Impulse Withstand Voltage | | kV | 4 | | | | | |
| Withstand Overload Current Capacity | | A | 8Ie (AC-7b), 10s | | | | | |
| Rated Insulation Voltage | | V | 500 | | | | | |
| Rated Operational Voltage | | V | 250V (1P/2P), 400V (3P/4P) | | | | | |
| Main Contact Type | | 1P | 10(1NO),01(1NC) | | | | | |
| | | 2P | 20(2NO),11(1NO+1NC),02(2NC) | | | | | |
| | | 3P | 30(3NO) 03(3NC) | | | | | |
| | | 4P | 40(4NO),22(2NO+2NC),31(3NO+1NC) | | | | | |
| Rated Operation Power | AC-7a | 230V kW | 3.5 | 4.5 | 5.5 | 7.2 | 9 | 14 |
| | | 400V kW | 6.3 | 10 | 16 | 21 | 26 | 40 |
| | AC-7b | 230V kW | 1.4 | 1.6 | 2 | 2.8 | 3.7 | 5 |
| | | 400V kW | 2.4 | 2.8 | 4 | 8.5 | 11 | 15 |
| Connection Screw | | | M3.5 | | | M5 | | |
| Tightening Torque | | N·m | 0.8 | | | 3.5 | | |
| Connection Wires | | $mm^2$ | 1.5-4 | | | 6-16 | | |
| Mechanical Life×$10^4$ | | Ops | 600 | | | | | |
| Electrical Life×$10^4$ | | Ops | 10 | | | | | |
| Rated Control Power Supply Voltage | | V | AC (50/60Hz): 24/48/110/230 DC: 12/24 | | | | | |
| Rated Lmiting Short Circuit Current | | kA | 3 | | | | | |
| Matching SCPD Type | | | 1P C16 | 1P C20 | 1P C25 | 1P C32 | 1P C40 | 1P C63 |



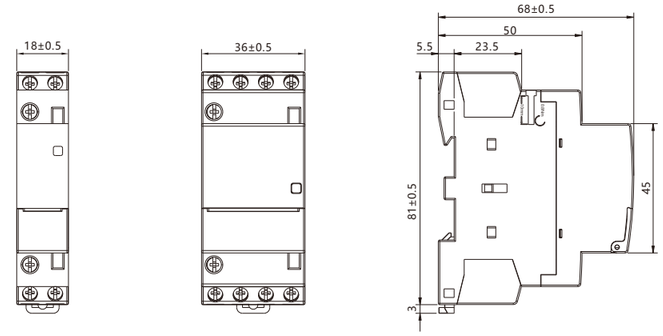




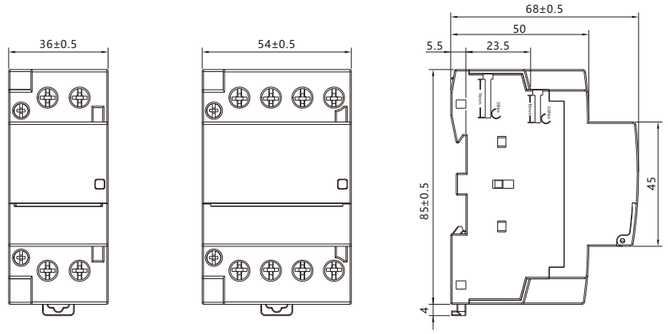
## Dimension

mm

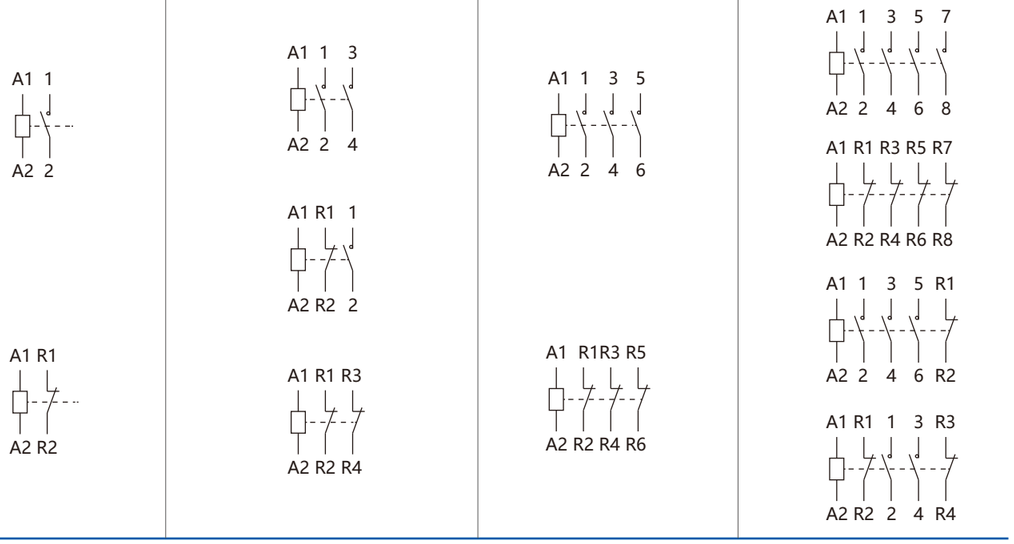
HCH8-16/20/25A

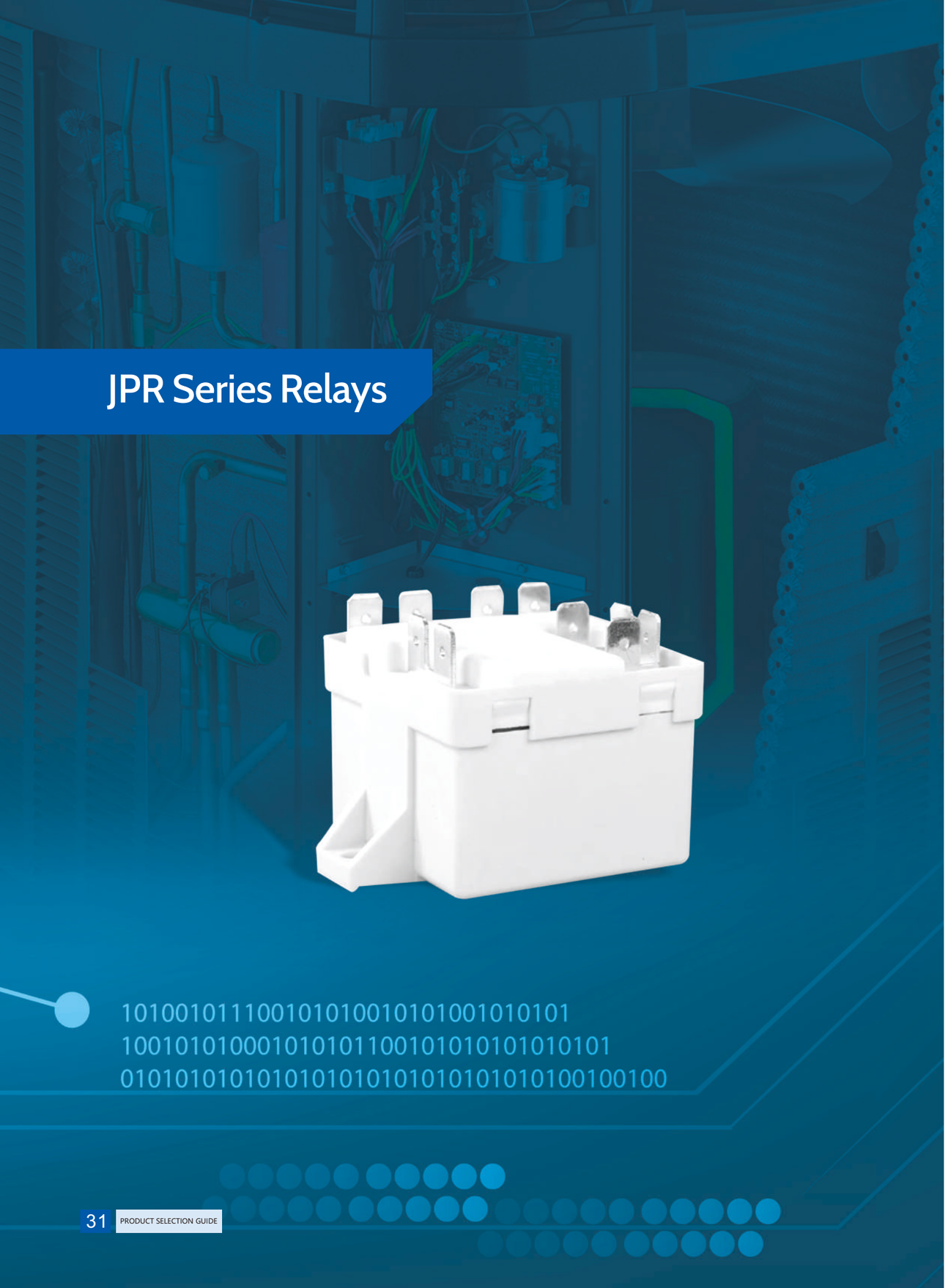



HCH8-32/40/63A




## Diagram








Hvacstar

## Product overview

Potential relay is designed to start single-phase motors which employ both start and run capacitor(CSR and CSIR configurations) for high starting torque.relays are manufactured for the HVAC industry. These relays meet all requirements for original equipment relays.

## Features&benefits

- ◇ 50A switching capability
- ◇ Has a variety of mounting positions
- ◇ With a variety of wiring
- ◇ Dust cover type packaging structure
- ◇ Used on heat pumps,vendingmachines,fans,and heating and heating and air conditioning



## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Code explanation

| | JPR | 550/ | AA | 1 | (XXX) |
|---|---|---|---|---|---|
| Type | | | | | |
| Coil number<br>549/550/554/555/558/556/559/560/561 | | | | | |
| Operation characteristics<br>AA to AS (See table for operating characteristics) | | | | | |
| Operation position<br>1/2/3/4/5/6 | | | | | |
| Special code1<br>XXX: Customer special requirement | | | | | |

## Product model and specification

### OPERATING CHARACTERISTICS AT 60Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

| Coil number | | 549 | | 550 | | 554 | | 555 | | 558 | | 556 | | 559 | | 560 | | 561 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vmax at 40°C (VAC) | | 170 | | 332 | | 336 | | 375 | | 379 | | 395 | | 420 | | 450 | | 500 | |
| Resistance at 25°C (Ω) ±10% | | 1500 | | 5600 | | 5700 | | 7500 | | 7600 | | 10000 | | 10700 | | 11000 | | 13800 | |
| | H.P.U. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. |
| AA | 120-130 | 111-124 | 20-45 | | | | | | | | | | | | 20-45 | | | | |
| AB | 130-140 | 120-134 | 20-45 | 132-148 | 40-90 | | | | | | | | | | | | | | |
| AC | 150-160 | 130-144 | 20-45 | 140-153 | 40-90 | | | | | | | | | | | | | | |
| AD | 160-170 | 140-153 | 20-45 | 150-163 | 40-90 | | | | | | | | | | | | | | |
| AE | 170-180 | 149-163 | 20-45 | 162-175 | 40-90 | | | | | | | | | | | | | | |
| AF | 180-190 | | | 171-184 | 40-90 | 162-175 | 40-90 | | | | | 180-195 | 40-105 | | | | | | |
| AG | 190-200 | | | 180-193 | 40-90 | 171-184 | 40-90 | 180-195 | 40-105 | | | 189-205 | 40-105 | 180-195 | 60-121 | | | | |
| AH | 200-220 | | | 186-215 | 40-90 | 180-193 | 40-90 | 190-215 | 40-105 | 180-195 | 40-105 | 186-214 | 60-130 | 189-224 | 60-121 | | | | |
| AI | 220-240 | | | 205-234 | 40-90 | 186-215 | 40-90 | 208-239 | 50-110 | 190-215 | 40-105 | 204-233 | 60-130 | 204-233 | 60-121 | | | | |
| AL | 240-260 | | | 224-252 | 40-105 | 205-234 | 40-90 | 224-252 | 50-110 | 208-239 | 50-110 | 223-252 | 60-130 | 223-259 | 60-121 | 223-252 | 60-130 | | |
| AM | 260-280 | | | 243-271 | 40-105 | 224-252 | 40-105 | 239-270 | 50-110 | 224-252 | 50-110 | 242-272 | 60-140 | 242-272 | 60-121 | 239-268 | 60-135 | 239-268 | 60-135 |
| AN | 280-300 | | | | | 243-271 | 40-105 | 260-289 | 50-110 | 239-270 | 50-110 | 262-290 | 60-140 | 262-290 | 60-121 | 258-287 | 60-135 | 258-287 | 60-135 |
| AO | 300-320 | | | | | | | | | 260-289 | 50-110 | 280-310 | 60-140 | 280-310 | 60-121 | 277-305 | 60-135 | 277-305 | 60-135 |
| AP | 320-340 | | | | | | | | | | | 300-328 | 60-140 | 300-328 | 60-121 | 295-324 | 60-135 | 295-324 | 60-135 |
| AQ | 340-360 | | | | | | | | | | | | | 318-347 | 60-121 | 314-342 | 60-135 | 314-342 | 60-135 |
| AR | 350-370 | | | | | | | | | | | | | | | | | 323-352 | 60-135 |
| AS | 360-380 | | | | | | | | | | | | | | | | | | |







### OPERATING CHARACTERISTICS AT 50Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

| Coil number | | 549 | | 550 | | 554 | | 555 | | 558 | | 556 | | 559 | | 560 | | 561 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vmax at 40°C (VAC) | | 151 | | 299 | | 300 | | 338 | | 340 | | 356 | | 379 | | 400 | | 450 | |
| Resistance at 25°C (Ω) ±10% | | 1500 | | 5600 | | 5700 | | 7500 | | 7600 | | 10000 | | 10700 | | 11000 | | 13800 | |
| | H.P.U. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. |
| A | 120-130 | 111-124 | 20-45 | | | | | | | | | | | | | | | | |
| B | 130-140 | 120-134 | 20-45 | 132-148 | 40-90 | | | | | | | | | | | | | | |
| C | 150-160 | 130-144 | 20-45 | 140-153 | 40-90 | | | | | | | | | | | | | | |
| D | 160-170 | 140-153 | 20-45 | 150-163 | 40-90 | | | | | | | | | | | | | | |
| E | 170-180 | 149-163 | 20-45 | 162-175 | 40-90 | | | | | | | | | | | | | | |
| F | 180-190 | | | 171-184 | 40-90 | 162-175 | 40-90 | | | | | 180-195 | 40-105 | | | | | | |
| G | 190-200 | | | 180-193 | 40-90 | 171-184 | 40-90 | 180-195 | 40-105 | | | 189-205 | 40-105 | 180-195 | 60-121 | | | | |
| H | 200-220 | | | 186-215 | 40-90 | 180-193 | 40-90 | 190-215 | 40-105 | 180-195 | 40-105 | 186-214 | 60-130 | 189-224 | 60-121 | | | | |
| I | 220-240 | | | 205-234 | 40-90 | 186-215 | 40-90 | 208-239 | 50-110 | 190-215 | 40-105 | 204-233 | 60-130 | 204-233 | 60-121 | | | | |
| L | 240-260 | | | 224-252 | 40-105 | 205-234 | 40-90 | 224-252 | 50-110 | 208-239 | 50-110 | 223-252 | 60-130 | 223-259 | 60-121 | 223-252 | 60-130 | | |
| M | 260-280 | | | 243-271 | 40-105 | 224-252 | 40-105 | 239-270 | 50-110 | 224-252 | 50-110 | 242-272 | 60-140 | 242-272 | 60-121 | 239-268 | 60-135 | 239-268 | 60-135 |
| N | 280-300 | | | | | 243-271 | 40-105 | 260-289 | 50-110 | 239-270 | 50-110 | 262-290 | 60-140 | 262-290 | 60-121 | 258-287 | 60-135 | 258-287 | 60-135 |
| O | 300-320 | | | | | | | | | 260-289 | 50-110 | 280-310 | 60-140 | 280-310 | 60-121 | 277-305 | 60-135 | 277-305 | 60-135 |
| P | 320-340 | | | | | | | | | | | 300-328 | 60-140 | 300-328 | 60-121 | 295-324 | 60-135 | 295-324 | 60-135 |
| Q | 340-360 | | | | | | | | | | | | | 318-347 | 60-121 | 314-342 | 60-135 | 314-342 | 60-135 |
| R | 350-370 | | | | | | | | | | | | | | | | | 323-352 | 60-135 |
| S | 360-380 | | | | | | | | | | | | | | | | | | |

If a 60Hz model is used at 50Hz, its 50Hz PU will be about 13% lower.
If a 50Hz model is used at 60Hz, its 60Hz PU will be about 13% higher

## Product Characteristics

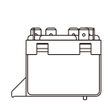
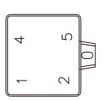
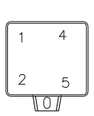
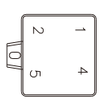
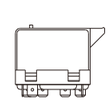
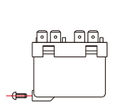
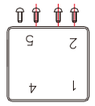
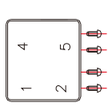
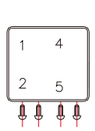
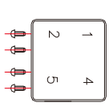
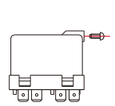
| | | |
|---|---|---|
| Maximum operating voltage | | 500VAC @-40℃~50℃ |
| | | 500VAC @-40℃~85℃ |
| Contact form | | SPST-NC |
| Contact resistance | | ≤50mΩ(@24V/1A) |
| Rated load | | 16A 400VAC |
| Maximum commutation voltage | | 500VAC |
| Maximum commutation current | | 50A |
| Input power | Holding power | ≤5 VA |
| | Input power | ≤21.5 VA |
| Mechanical durability | | $1X10^6$ |
| Electrical durability | | SPST-NC: 5 x $10^5$OPS 16A on and off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off<br>SPST-NC: 2 x $10^5$OPS 35A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off<br>SPST-NC: 1 x $10^5$OPS 50A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off |
| Insulation resistance | | 500MΩ（500VDC） |
| Shock（stability） | | $98m/s^2$ |
| Vibration | | 10Hz to 55Hz, 1.5mm double amplitude |
| Humidity range | | 5% to 85%RH |
| Temperature range | | -40℃ to 85℃ |
| Lead out form | | Quick connect |
| Weight | | About 98g |
| Packaging method | | Dust cover type |





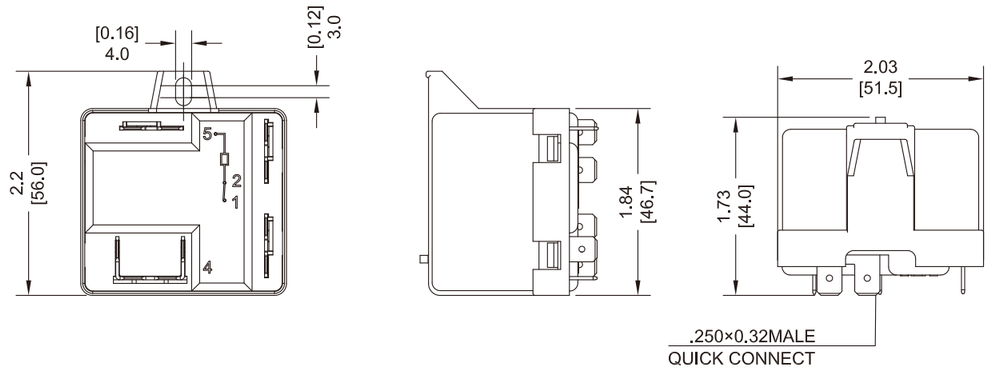


## Operating position

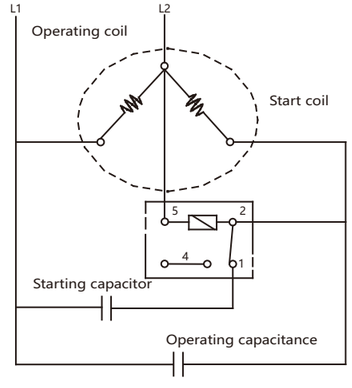
| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| TAB MOUNT |  |  |  |  |  |  |
| PANEL MOUNT |  |  |  |  |  |  |

## Dimensions

inch[mm]




## Wiring diagram



# PR90 Series Relays






## Product overview

refrigeration potential relay service kit has become an industry standard since its introduction over fifty years ago. Its long-term reliability and easy replacement have been the key toits success. The wide calibration ranges and the universal bracket make potential relay is a must on every service truck.



## Features&benefits

- ◇ 50A switching capability
- ◇ Has a variety of mounting positions
- ◇ With a variety of wiring
- ◇ Dust cover type packaging structure
- ◇ Used on heat pumps,vendingmachines,fans,and heating and heating and air conditioning

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Code explanation

| | | | | | | |
|---|---|---|---|---|---|---|
| | PR-90 | 60/ | Z | AA | 1 | (XXX) |

Type

Coil number
63/50/65/69/58/66/71/61/64

Terminal configuration
Z Metal Tab Mount
D Plastic Tab Mount
P Panel Mount

Operation characteristics AA to AS
(See table for operating characteristics)

Operation position
1/2/3/4/5/6

Special code1
XXX: Customer special requirement

## Product model and specification

### OPERATING CHARACTERISTICS AT 60Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

| Coil number | | 63 | | 50 | | 65 | | 69 | | 58 | | 66 | | 71 | | 61 | | 64 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vmax at 40°C (VAC) | | 170 | | 332 | | 336 | | 375 | | 379 | | 395 | | 420 | | 450 | | 500 | |
| Resistance at 25°C (Ω) ±10% | | 1500 | | 5600 | | 5700 | | 7500 | | 7600 | | 10000 | | 10700 | | 11000 | | 13800 | |
| | H.P.U. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. |
| AA | 120-130 | 111-124 | 20-45 | | | | | | | | | | | | | | | | |
| AB | 130-140 | 120-134 | 20-45 | 132-148 | 40-90 | | | | | | | | | | | | | | |
| AC | 150-160 | 130-144 | 20-45 | 140-153 | 40-90 | | | | | | | | | | | | | | |
| AD | 160-170 | 140-153 | 20-45 | 150-163 | 40-90 | | | | | | | | | | | | | | |
| AE | 170-180 | 149-163 | 20-45 | 162-175 | 40-90 | | | | | | | | | | | | | | |
| AF | 180-190 | | | 171-184 | 40-90 | 162-175 | 40-90 | | | | | 180-195 | 40-105 | | | | | | |
| AG | 190-200 | | | 180-193 | 40-90 | 171-184 | 40-90 | 180-195 | 40-105 | | | 189-205 | 40-105 | 180-195 | 60-121 | | | | |
| AH | 200-220 | | | 186-215 | 40-90 | 180-193 | 40-90 | 190-215 | 40-105 | 180-195 | 40-105 | 186-214 | 60-130 | 189-224 | 60-121 | | | | |
| AI | 220-240 | | | 205-234 | 40-90 | 186-215 | 40-90 | 208-239 | 50-110 | 190-215 | 40-105 | 204-233 | 60-130 | 204-233 | 60-121 | | | | |
| AL | 240-260 | | | 224-252 | 40-105 | 205-234 | 40-90 | 224-252 | 50-110 | 208-239 | 50-110 | 223-252 | 60-130 | 223-259 | 60-121 | 223-252 | 60-130 | | |
| AM | 260-280 | | | 243-271 | 40-105 | 224-252 | 40-105 | 239-270 | 50-110 | 224-252 | 50-110 | 242-272 | 60-140 | 242-272 | 60-121 | 239-268 | 60-135 | 239-268 | 60-135 |
| AN | 280-300 | | | | | 243-271 | 40-105 | 260-289 | 50-110 | 239-270 | 50-110 | 262-290 | 60-140 | 262-290 | 60-121 | 258-287 | 60-135 | 258-287 | 60-135 |
| AO | 300-320 | | | | | | | | | 260-289 | 50-110 | 280-310 | 60-140 | 280-310 | 60-121 | 277-305 | 60-135 | 277-305 | 60-135 |
| AP | 320-340 | | | | | | | | | | | 300-328 | 60-140 | 300-328 | 60-121 | 295-324 | 60-135 | 295-324 | 60-135 |
| AQ | 340-360 | | | | | | | | | | | | | 318-347 | 60-121 | 314-342 | 60-135 | 314-342 | 60-135 |
| AR | 350-370 | | | | | | | | | | | | | | | | | 323-352 | 60-135 |
| AS | 360-380 | | | | | | | | | | | | | | | | | | |



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PR90 Series Relays

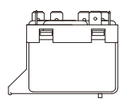
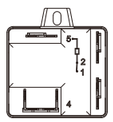
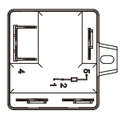
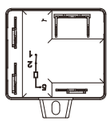
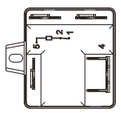
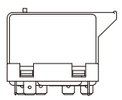
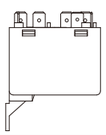
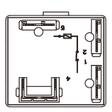
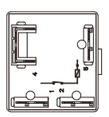
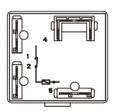
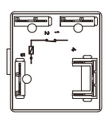
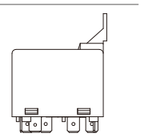
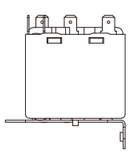
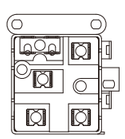
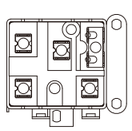
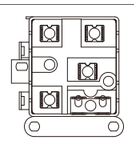
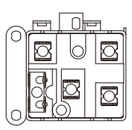
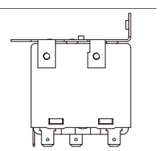
### OPERATING CHARACTERISTICS AT 50Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

| Coil number | | 63 | | 50 | | 65 | | 69 | | 58 | | 66 | | 71 | | 61 | | 64 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vmax at 40°C (VAC) | | 151 | | 299 | | 300 | | 338 | | 340 | | 356 | | 379 | | 400 | | 450 | |
| Resistance at 25°C (Ω) ±10% | | 1500 | | 5600 | | 5700 | | 7500 | | 7600 | | 10000 | | 10700 | | 11000 | | 13800 | |
| | H.P.U. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. | P.U. | D.O. |
| A | 120-130 | 111-124 | 20-45 | | | | | | | | | | | | | | | | |
| B | 130-140 | 120-134 | 20-45 | 132-148 | 40-90 | | | | | | | | | | | | | | |
| C | 150-160 | 130-144 | 20-45 | 140-153 | 40-90 | | | | | | | | | | | | | | |
| D | 160-170 | 140-153 | 20-45 | 150-163 | 40-90 | | | | | | | | | | | | | | |
| E | 170-180 | 149-163 | 20-45 | 162-175 | 40-90 | | | | | | | | | | | | | | |
| F | 180-190 | | | 171-184 | 40-90 | 162-175 | 40-90 | | | | | 180-195 | 40-105 | | | | | | |
| G | 190-200 | | | 180-193 | 40-90 | 171-184 | 40-90 | 180-195 | 40-105 | | | 189-205 | 40-105 | 180-195 | 60-121 | | | | |
| H | 200-220 | | | 186-215 | 40-90 | 180-193 | 40-90 | 190-215 | 40-105 | 180-195 | 40-105 | 186-214 | 60-130 | 189-224 | 60-121 | | | | |
| I | 220-240 | | | 205-234 | 40-90 | 186-215 | 40-90 | 208-239 | 50-110 | 190-215 | 40-105 | 204-233 | 60-130 | 204-233 | 60-121 | | | | |
| L | 240-260 | | | 224-252 | 40-105 | 205-234 | 40-90 | 224-252 | 50-110 | 208-239 | 50-110 | 223-252 | 60-130 | 223-259 | 60-121 | 223-252 | 60-130 | | |
| M | 260-280 | | | 243-271 | 40-105 | 224-252 | 40-105 | 239-270 | 50-110 | 224-252 | 50-110 | 242-272 | 60-140 | 242-272 | 60-121 | 239-268 | 60-135 | 239-268 | 60-135 |
| N | 280-300 | | | | | 243-271 | 40-105 | 260-289 | 50-110 | 239-270 | 50-110 | 262-290 | 60-140 | 262-290 | 60-121 | 258-287 | 60-135 | 258-287 | 60-135 |
| O | 300-320 | | | | | | | | | 260-289 | 50-110 | 280-310 | 60-140 | 280-310 | 60-121 | 277-305 | 60-135 | 277-305 | 60-135 |
| P | 320-340 | | | | | | | | | | | 300-328 | 60-140 | 300-328 | 60-121 | 295-324 | 60-135 | 295-324 | 60-135 |
| Q | 340-360 | | | | | | | | | | | | | 318-347 | 60-121 | 314-342 | 60-135 | 314-342 | 60-135 |
| R | 350-370 | | | | | | | | | | | | | | | | | 323-352 | 60-135 |
| S | 360-380 | | | | | | | | | | | | | | | | | | |

If a 60Hz model is used at 50Hz, its 50Hz PU will be about 13% lower.
If a 50Hz model is used at 60Hz, its 60Hz PU will be about 13% higher.

## Operating position

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| PLASTIC TAB MOUNT |  |  |  |  |  |  |
| PANEL MOUNT |  |  |  |  |  |  |
| METAL TAB MOUNT |  |  |  |  |  |  |

## Product Characteristics

| | | |
|---|---|---|
| Maximum operating voltage | | 500VAC @-40℃~50℃ |
| | | 500VAC @-40℃~85℃ |
| Contact form | | SPST-NC |
| Contact resistance | | ≤50mΩ(@24V/1A) |
| Rated load | | 35A 400VAC |
| Maximum commutation voltage | | 500VAC |
| Maximum commutation current | | 50A |
| Input power | Holding power | ≤5 VA |
| | Input power | ≤21.5 VA |
| Mechanical durability | | $1X10^6$ |
| Electrical durability | | SPST-NC: 5 x $10^5$OPS 16A on and off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off<br>SPST-NC: 2 x $10^5$OPS 35A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off<br>SPST-NC: 1 x $10^5$OPS 50A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off |
| Insulation resistance | | 500MΩ（500VDC） |
| Shock（stability） | | $98m/s^2$ |
| Vibration | | 10Hz to 55Hz, 1.5mm double amplitude |
| Humidity range | | 5% to 85%RH |
| Temperature range | | -40℃ to 85℃ |
| Lead out form | | Quick connect |
| Weight | | About 98g |
| Packaging method | | Dust cover type |



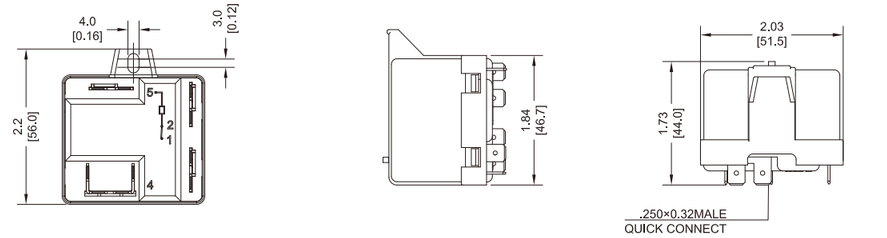




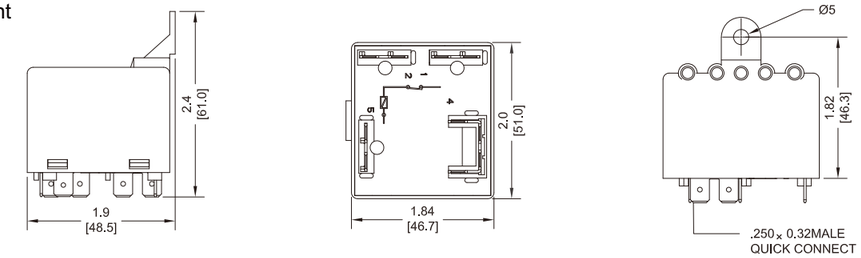
## Dimensions

inch[mm]

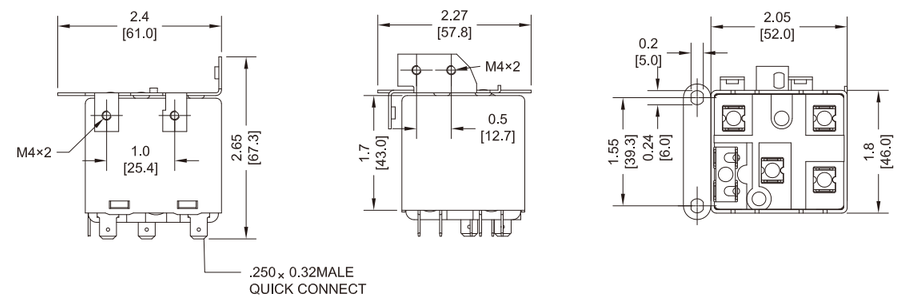
Plastic Tab Mount



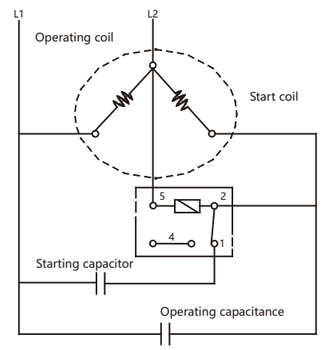

Panel Mount




Metal Tab Mount




## Wiring diagram



# PR3 Series Relays






Hvacstar



## Product overview

Relays control circuits through a low-power signal.An electro-magnetic current actuates the relay, creating a closed circuit within the device. General purpose relays are used in applications including HVAC systems, standby power supplies and telephone exchanges. We carry extensive inventory to meet delivery schedules.

## Features&benefits

- ◇ 25A switching capability
- ◇ Has a variety of mounting positions
- ◇ With a variety of wiring
- ◇ Dust cover type packaging structure
- ◇ Double break contact

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Product model and specification

| Item ID | Rated voltage | Pick-Up | Drop Out | Coil resistance | Contact resistance | Contact form |
|---|---|---|---|---|---|---|
| | VAC | VAC | VAC | Ω | mΩ | |
| PR360 | 24 | ≤20.4 | ≥4.8 | 85×(1±10%) | ≤100 | SPST-NO |
| PR362 | 120 | ≤102.0 | ≥24 | 2000×(1±10%) | ≤100 | SPST-NO |
| PR364 | 240 | ≤204.0 | ≥48 | 7200×(1±10%) | ≤100 | SPST-NO |
| PR366 | 277 | ≤235 | ≥55.7 | 11000×(1±10%) | ≤100 | SPST-NO |
| PR370 | 24 | ≤20.4 | ≥4.8 | 90×(1±10%) | ≤100 | SPDT |
| PR372 | 120 | ≤102.0 | ≥24 | 2000×(1±10%) | ≤100 | SPDT |
| PR374 | 240 | ≤204.0 | ≥48 | 7200×(1±10%) | ≤100 | SPDT |
| PR376 | 277 | ≤235 | ≥55.7 | 11000×(1±10%) | ≤100 | SPDT |
| PR380 | 24 | ≤20.4 | ≥4.8 | 90×(1±10%) | ≤100 | SPST NO+NC |
| PR382 | 120 | ≤102.0 | ≥24 | 2000×(1±10%) | ≤100 | SPST NO+NC |
| PR384 | 240 | ≤204.0 | ≥48 | 7200×(1±10%) | ≤100 | SPST NO+NC |

## Product Characteristics

| | | |
|---|---|---|
| Contact form | | SPST-NO/SPDT/SPST NO+NC |
| Contact resistance | | ≤100mΩ（@24V/1A） |
| Rated load | | 25A 277VAC (breaking only) |
| Maximum commutation voltage | | 277VAC |
| Maximum commutation current | | 25A |
| Maximum switching power | | 4000VA |
| Mechanical durability | | $1X10^6$ |
| Electrical durability | | $5X10^5$ 25A 277VAC, resistive load, 65 ℃, 1s on and 9s off<br>$3X10^4$ 3A 277VAC, general load, 65 ℃, 1s on and 9s off |
| Insulation resistance | | 500MΩ（500VDC） |
| Dielectric Strength | Between coil and contact | 2500VAC 1min |
| | Between disconnecting contacts | 1000VAC 1min |
| Action time | | ≤25ms |
| Release time | | ≤25ms |
| Shock（stability） | | $98m/s^2$ |
| Vibration | | 10Hz to 55Hz, 1.5mm double amplitude |
| Humidity range | | 5% - 85%RH |
| Temperature range | | -40℃ - 85℃ |
| Lead out form | | Quick connect |
| Weight | | About 75g |
| Packaging method | | Dust cover type |

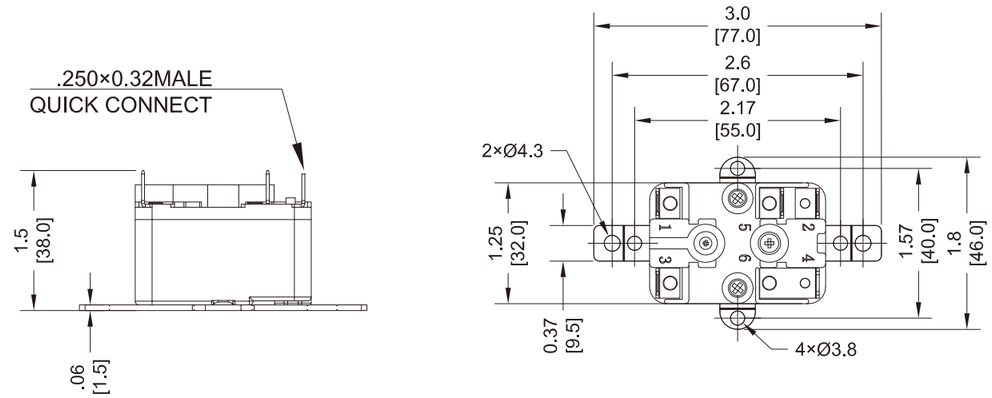




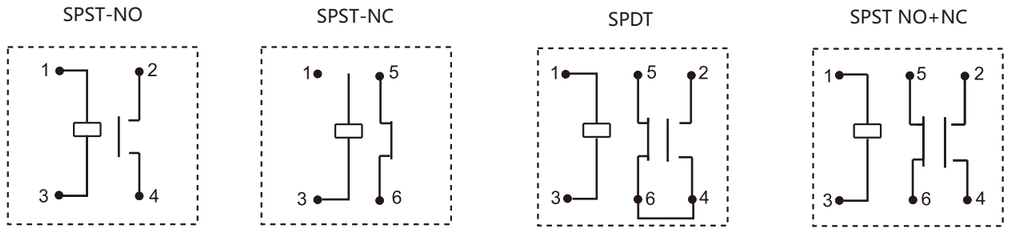


## Dimensions

inch[mm]




## Wiring diagram



# PR29 Series Relays






Hvacstar



## Product overview

Relays control circuits through a low-power signal.An electro-magnetic current actuates the relay, creating a closed circuit within the device. General purpose relays are used in applications including HVAC systems, standby power supplies and telephone exchanges. We carry extensive inventory to meet delivery schedules.

## Features&benefits

- ◇16A switching capability
- ◇Has a variety of mounting positions
- ◇With a variety of wiring
- ◇Dust cover type packaging structure

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Product model and specification

| Item ID | Rated voltage | Pick-Up | Drop Out | Coil resistance | Contact resistance | Contact form |
|---|---|---|---|---|---|---|
| | VAC | VAC | VAC | Ω | mΩ | |
| PR290Q | 24 | ≤20.4 | ≥4.8 | 80×(1±10%) | ≤50 | SPST-NO |
| PR291Q | 120 | ≤102.0 | ≥24 | 1900×(1±10%) | ≤50 | SPST-NO |
| PR292Q | 240 | ≤204.0 | ≥48 | 7200×(1±10%) | ≤50 | SPST-NO |
| PR293Q | 24 | ≤20.4 | ≥4.8 | 90×(1±10%) | ≤50 | SPDT |
| PR294Q | 120 | ≤102.0 | ≥24 | 2000×(1±10%) | ≤50 | SPDT |
| PR295Q | 240 | ≤204.0 | ≥48 | 7200×(1±10%) | ≤50 | SPDT |

## Product Characteristics

| | | |
|---|---|---|
| Contact form | | SPST-NO/ SPST-NC /SPDT |
| Contact resistance | | ≤50mΩ (@24V/1A) |
| Rated load | | 16A 277VAC (breaking only) |
| Maximum commutation voltage | | 277VAC |
| Maximum commutation current | | 16A |
| Maximum switching power | | 4000VA |
| Mechanical durability | | $1X10^6$ |
| Electrical durability | | $1X10^5$ 16A 250VAC , General load, 65 ℃, 1s on and 9s off<br>$3X10^4$ 8A 250VAC, General load, 40 ℃, 1s on and 9s off<br>$1X10^5$ 16A 250VAC, General load, 65 ℃, ls on and 9s off |
| Insulation resistance | | 500MΩ (500VDC) |
| Dielectric Strength | Between coil and contact | 2500VAC 1min |
| | Between disconnecting contacts | 1000VAC 1min |
| Action time: | | ≤25ms |
| Release time: | | ≤25ms |
| Shock (stability) | | $98m/s^2$ |
| Vibration | | 10Hz to 55Hz, 1.5mm double amplitude |
| Humidity range | | 5% - 85%RH |
| Temperature range | | -40℃ - 85℃ |
| Lead out form | | Quick connect |
| Weight | | About 74g |
| Packaging method | | Dust cover type |

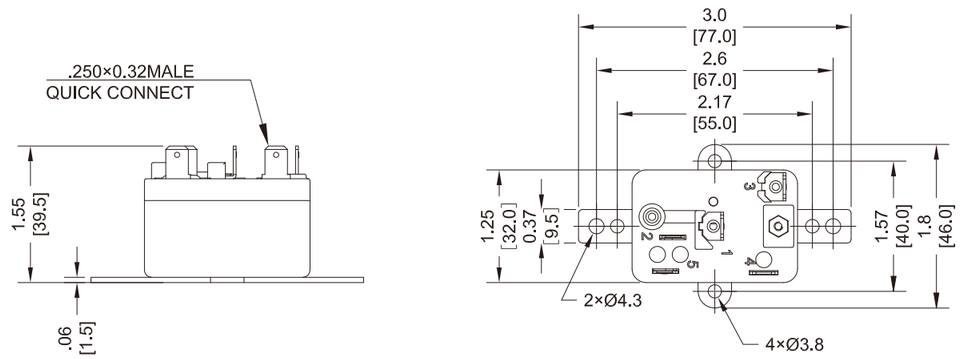






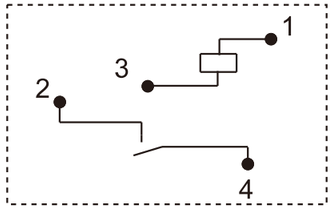
## Dimensions

inch[mm]

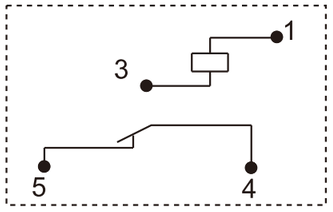



## Wiring diagram

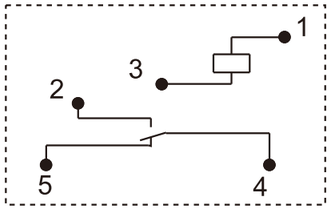
SPST-NO




SPST-NC




SPDT








Hvacstar

## Product overview

Relays control circuits through a low-power signal.An electro-magnetic current actuates the relay, creating a closed circuit within the device. General purpose relays are used in applications including HVAC systems, standby power supplies and telephone exchanges.
We carry extensive inventory to meet delivery schedules.



## Features&benefits

- ◇ 25A switching capability
- ◇ Has a variety of mounting positions
- ◇ With a variety of wiring
- ◇ Dust cover type packaging structure
- ◇ Used on heat pumps, vending machines, fans, and heating and heating and air conditioning

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Product model and specification

| Item ID | Rated voltage | Pick-Up | Drop Out | Coil resistance | Contact resistance | Contact form |
|---|---|---|---|---|---|---|
| | VAC | VAC | VAC | Ω | mΩ | |
| PR-340 | 24 | ≤20.4 | ≥4.8 | 21×(1±10%) | ≤50 | DPDT |
| PR-341 | 104/120 | ≤102.0 | ≥24 | 440×(1±10%) | ≤50 | DPDT |
| PR-342 | 208/240 | ≤204.0 | ≥48 | 1600×(1±10%) | ≤50 | DPDT |
| PR-343 | 24 | ≤20.4 | ≥4.8 | 21×(1±10%) | ≤50 | SPDT |
| PR-344 | 104/120 | ≤102.0 | ≥24 | 440×(1±10%) | ≤50 | SPDT |
| PR-345 | 208/240 | ≤204.0 | ≥48 | 1600×(1±10%) | ≤50 | SPDT |
| PR-346 | 24 | ≤20.4 | ≥4.8 | 21×(1±10%) | ≤50 | DPST-NO |
| PR-347 | 104/120 | ≤102.0 | ≥24 | 440×(1±10%) | ≤50 | DPST-NO |
| PR-348 | 208/240 | ≤204.0 | ≥48 | 1600×(1±10%) | ≤50 | DPST-NO |

## Product Characteristics

| | | |
|---|---|---|
| Dontact form | | SPST-NO/SPST-NC/ SPDT<br>DPST-NO/DPST-NC/DPDT |
| Dontact resistance | | ≤50mΩ （@24V/1A） |
| | | |
| Maximum commutation voltage | | 277VAC |
| Maximum commutation current | | 25A |
| Input power | Holding power | 9.5 VA |
| | Input power | 21.5 VA |
| Contact rating | | 12 FLA, 60 LRA, 15A resistive @ 125VAC;<br>6 FLA, 35 LRA, 15A resistive @ 250/277VAC;<br>3 FLA, 18 LRA, 12.5A resistive @ 480VAC;<br>3 FLA, 14 LRA @ 600VAC; |
| Mechanical durability | | $1\times10^6$ |
| Electrical durability | | $2.5\times10^5$ operations., at rated resistive loads.<br>$1.0\times10^5$ operations., at rated inductive loads. |
| Insulation resistance | | 500MΩ（500VDC） |
| Dielectric Strength | Between coil and contact | 2200VAC 1min |
| | Between disconnecting contacts | 1000VAC 1min |
| Action time: | | ≤25ms |
| Release time: | | ≤25ms |
| Shock（stability） | | $98m/s^2$ |
| Vibration | | 10Hz to 55Hz, 1.5mm double amplitude |
| Humidity range | | 5% - 85%RH |
| Temperature range | | -40℃ - 65℃ |
| Lead out form | | Quick connect |
| weight | | About 173g |
| Packaging method | | Dust cover type |

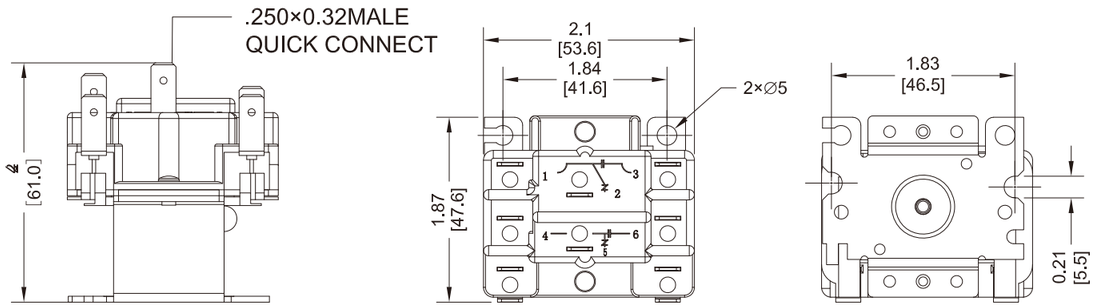




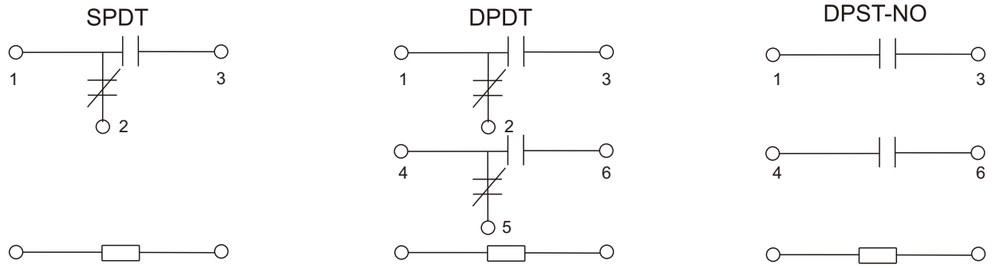


## Dimensions

inch[mm]




## Wiring diagram



# PR60 Series Relays






Hvacstar

## Product overview

The relay has improved conductivity and heat dissipation capabilities on key components that can be used in thousands of higher power and temperature environments, such as charging applications. The PR60 two-pole power relay is ideal for charging, HVAC, industrial control, power and lighting applications.



## Features&benefits

- ◇ DPST-NO DPST-NC and DPDT configuration
- ◇ Meets 8mm creepage, 4kV dielectric
- ◇ Epoxy sealed version available
- ◇ UL Class F (155°C) standard

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## Product Model

| | PR60 | 2A | 24 | D | E | F |
|---|---|---|---|---|---|---|

Type

Contact Form
2A：2 groups normally open
2B：2 groups normally close
2C：2 groups conversion

Coil Voltage:DC:5/6/12/24/48/110
AC:12/24/120/220/240/277

Voltage specification:
D: DC A5: AC 50Hz A6: AC 50/60Hz

Protection regulation
Nil: No seals E: seal

pin specification
F: 6.35×0.8mm KF: 4.75×0.5mm PF: PCB

## General Data

| | |
|---|---|
| Life Expectancy<br>Mechanical<br>Electrical | Minimum operations<br>$5\times10^6$<br>$1\times10^5$ at 30A, 277VAC Res. (N.0.) |
| Operate Time | 15ms typical<br>25ms maximum with bounce |
| Release Time | 10ms typical<br>25ms maximum with bounce<br>(with no coil suppression) |
| Dielectric Strength<br>(at sea level for 1 min.) | 1500Vrms contact to contact<br>4000Vrms contact to coil<br>2000Vrms between contact sets |
| Insulation Resistance | 1000MΩ (500VDC) |
| Dropout | DC:Greater than 10%of nominal coil voltage<br>AC: Greater than 20%of nominal coil voltage |
| Ambiant Temperature<br>Operating | At nominal coil voltage<br>DC: -40℃ ( -40°F) to 85℃ (185°F)<br>AC: -40℃ ( -40°F) to 65℃ (149°F)<br>-40℃ ( -40°F) to 105℃ (221°F) |
| Vibration | 0.062" (1.65mm) DA at 10-55Hz |
| Shock | Operational, 10g for 11ms 1/2 sine pulse<br>Wno contact opening > 100usec)<br>Non-destructive, 100g for 11ms 1/2 sine pulse |
| Enclosure | P.B.T. polyester |
| Terminals | Quick connect tabs<br>Note: Allow suitable slack on leads when wiring.<br>and do not subject the terminals to excessive force. |
| Max. Immersion Time | 30 seconds |
| Weight | 86 grams |
| Packing unit in pos | 20 per plastic tray / 100 per carton box |







## Contacts

| | |
|---|---|
| Arrangement | DPST(2 Form A) DPST(2 FormB)<br>DPDT(2 Form C) |
| Ratings | Resistive load:<br>Max. switched power: 1200 W or 11080 VA<br>Max. switched current: 40A (N.O), 3A (N.C.)<br>Max.switchedvoltage:30VDC*or600VAC<br><br>*Notertf switching voltage is greater than 30VDC,<br>special precautions must be taken. Please contact the factory. |
| Rated Load<br>UL | Normally open contacts (N.O.)<br>40A at 277VAC, Resistive, 30k cycles [2][3]<br>30A at 277VAC, General Use, 100k cycles [2][3]<br>10A at 600VAC, General Use, 6k cycles [1][3]<br>1HP at 120VAC, 50k cycles [1][3]<br>2HP at 240VAC, 50k cycles [1][3]<br>2HP at 600VAC, 100k cycles [1][3]<br><br>Normally open contacts (N.O.), DC Coils only<br>25.3FLA/ 110LRA at 240VAC, 100k cycles [1][3]<br><br>Normally closed contacts (N.C.)<br>3A at 277VAC, General Use, 100k cycles [2][3]<br>3A at 250VAC, General Use, 100k cycles [2][3]<br>3A at 240VAC, General Use, 100k cycles [2][3]<br>3A at 125VAC, General Use, 100k cycles [2][3]<br>3FLA/ 3LRA at 240VAC, 30k cycles [1][3] |
| Resistance | ≤50mΩ（@24V/1A) |

## Notes

1. All values at 40°C (104°F).
2. All values at 85°C (185°F).
3. Relay may pull in with less than "Must Operate" value.
4. Specifications subject to change without notice.

## Coil Specifications

### DC Coil

| Nominal Coil VDC | Must Operate VDC | Max. Continuous VDC | Nominal Current mA ± 10% | Coil Resistance Ohm ± 10% |
|---|---|---|---|---|
| 5 | 3.8 | 8.0 | 326.7 | 15.3 |
| 6 | 4.5 | 10.5 | 272.0 | 22 |
| 12 | 9.0 | 20.7 | 140.0 | 86 |
| 24 | 18.0 | 41.8 | 68.5 | 350 |
| 48 | 36.0 | 83.4 | 34.5 | 1390 |
| 110 | 82.5 | 190.5 | 15.2 | 7255 |

### AC Coil

| Nominal Coil VAC | Must Operate VAC | Max. Continuous VAC | Nominal Current mA ± 10% | 50Hz Coil Resistance Ohm ± 10% | 60Hz Coil Resistance Ohm ± 10% |
|---|---|---|---|---|---|
| 12 | 9.6 | 15.6 | 340.0 | 9.5 | 8 |
| 24 | 19.2 | 31.2 | 166.0 | 45 | 35.7 |
| 120 | 96.0 | 156.0 | 33.3 | 1125 | 830 |
| 220 | 176.0 | 286.0 | 18.2 | 3800 | 2870 |
| 240 | 192.0 | 312.0 | 16.7 | 4500 | 3800 |
| 277 | 221.6 | 360.1 | 14.4 | 5960 | 4700 |

### Power consumption

| | |
|---|---|
| At Pickup Voltage | 925mW DC coil |
| At Pickup Voltage | 2.6VA AC coil |
| Max. Continuous | 5.0W at 20℃ (68°F) ambient ,DC coil |
| Dissipation | 7.0VA at 20℃ (68°F) ambient, AC coil |
| Temperature Rise | 48°C(86°F) at nominal coil voltage, DC coil<br>68°C(122°F) at nominal coilvoltage, AC coil |
| Temperature | Max. 155℃ (311°F) |

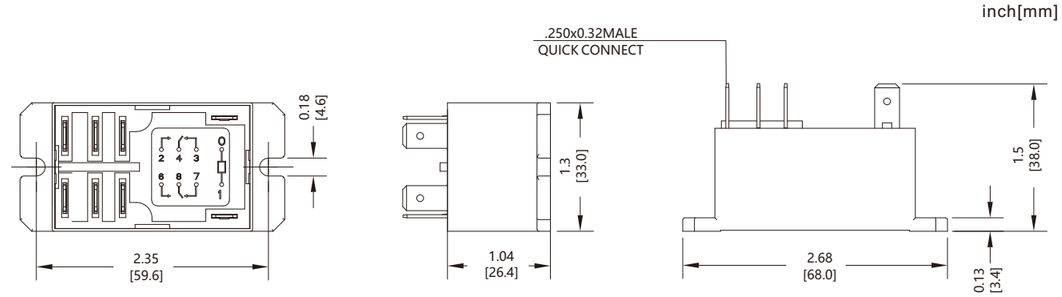


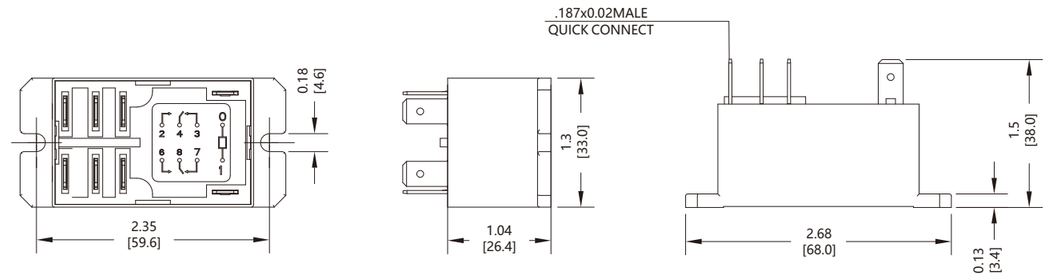
Hvacstar

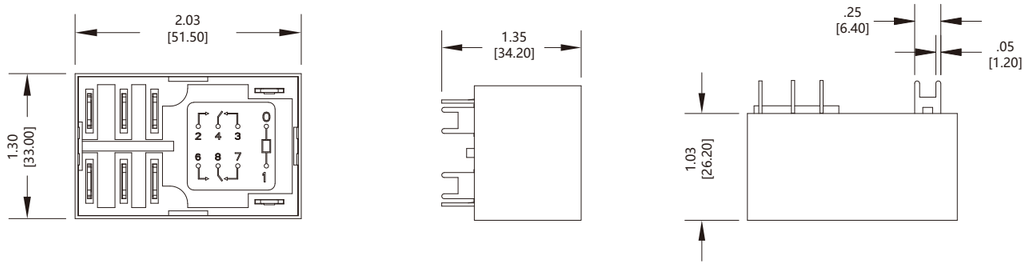
PR60 Series Relays

## Dimension

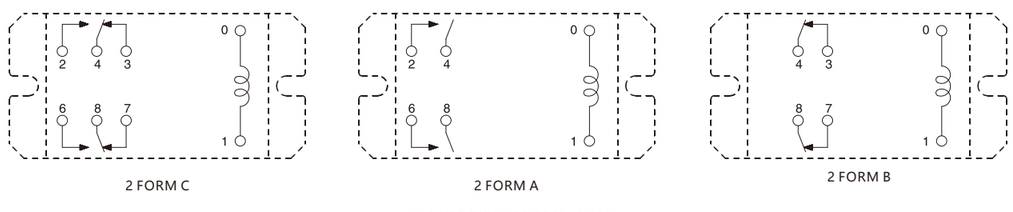
inch[mm]



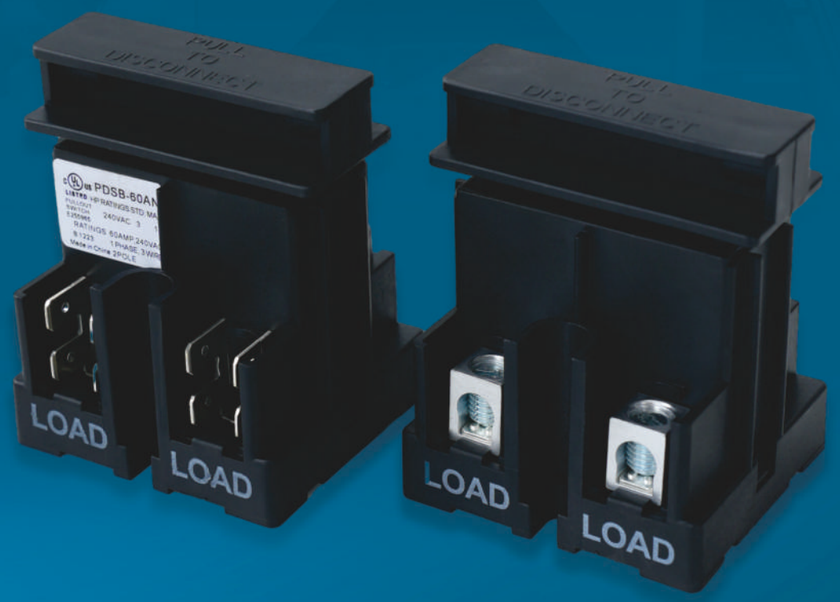




## Wiring Diagram



# PDSB Pull-out Disconnect






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## Product overview

The Most Widely Used Application for Air Conditioning Disconnects is for Residential and Light Commerical AirConditioning Units.An Air Conditioning Disconnects are also found in use with Heat Pumps, Swimming Pools,Spas, Whirlpools and Pump Houses and Required for Servicing Electric Water Heaters.



## Features & benefits

Box Lugs Quick-Connect available on Load side
Box Lugs standard on Line side
Available with or without gasket

## Product certification

| | | |
|---|---|---|
| c UL us LISTED | UL | UL 1429 |
| RoHS | RoHS | RoHS Compliant |

## Technical Data

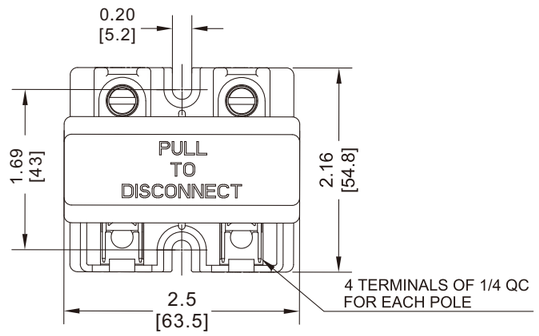
Pull out Disconnect Switch

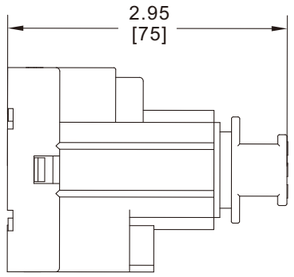
| Cat. No. | Description | Main Ampere Rating | HP Rating | Wire Size(AWG) & Torque Cu/AL |
|---|---|---|---|---|
| | | | 240V | LINE/LOAD |
| PDSB-60AN (LAOD 4QC) | Non-Enclosed Non-Fused Pull-Out Switch | 60AMP 240V 1 PHASE 2-Pole 3-WIRE | STD. 3 MAX. 10 | #14-10 35LB-IN<br>#8-6 45LB-IN<br>#4-2 50LB-IN |
| PDSB-60AN (LOAD LUG) | Non-Enclosed Non-Fused Pull-Out Switch | 60AMP 240V 1 PHASE 2-Pole 3-WIRE | STD. 3 MAX. 10 | #14-10 35LB-IN<br>#8-6 45LB-IN<br>#4-2 50LB-IN |

## Dimensions

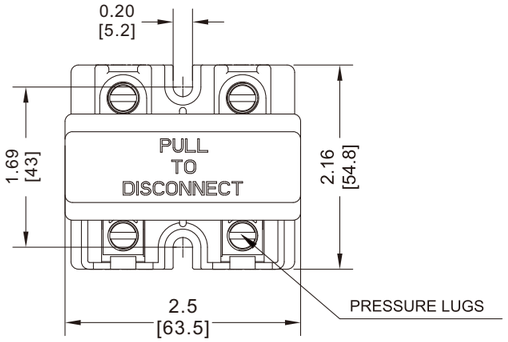
inch[mm]

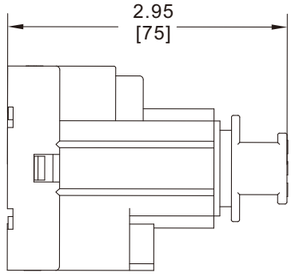
### PDSB-60AN(LOAD 4QC)




### PDSB-60AN(LOAD LUG)






# PDS&PCDS Air Conditioning Disconnects



# Product overview

The Most Widely Used Application for Air Conditioning Disconnects is for Residential and Light Commerical AirConditioning Units.An Air Conditioning Disconnects are also found in use with Heat Pumps, Swimming Pools,Spas, Whirlpools and Pump Houses and Required for Servicing Electric Water Heaters.

## Features & benefits

- ◇ Metallic and non-metallic options are NEMA 3R rated outdoor enclosures
- ◇ Easy-to-remove high-strength protective shield
- ◇ Easy-to-remove front cover
- ◇ Copper-rated line and load lugs
- ◇ Ample wiring space for mounting with a stud gun
- ◇ Padlock-able door provision for safety and reduction of tampering
- ◇ Factory-installed tamper resistant/weather-resistant ground fault receptacles

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 1429 |
| RoHS | RoHS | RoHS Compliant |

## Technical Data

Air Conditioning Disconnects

| Cat. No. | Description | Main Ampere Rating | HP Rating | | Wire Size(AWG) & Torque Cu/AL | |
|---|---|---|---|---|---|---|
| | | | 120V | 240V | LINE/LOAD | GROUND |
| PDS-30A<br>PCDS-30A | Enclosed Fused ACD | 30AMP 240V<br>1 PHASE 2-Pole 3-WIRE | STD. 0.5<br>MAX. 2 | STD. 1.5<br>MAX. 3 | #14-10 35LB-IN<br>#8-6 45LB-IN<br>#4-2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |
| PDS-60A<br>PCDS-60A | Enclosed Fused ACD | 60AMP 240V<br>1 PHASE 2-Pole 3-WIRE | STD. 1.5<br>MAX. 3 | STD. 3<br>MAX. 10 | #14-10 35LB-IN<br>#8-6 45LB-IN<br>#4-2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |
| PDS-60AN<br>PCDS-60AN<br>PCDS-60AN(Mini) | Enclosed Non-Fused ACD | 60AMP 240V<br>1 PHASE 2-Pole 3-WIRE | - | STD. 3<br>MAX. 10 | #14-10 35LB-IN<br>#8-6 45LB-IN<br>#4-2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |



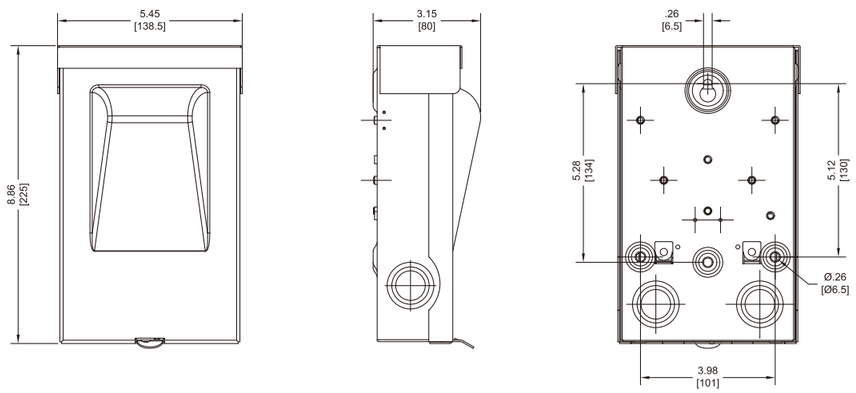




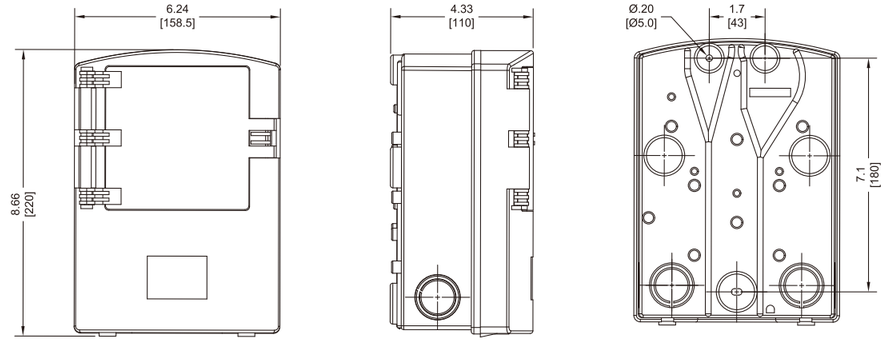
## Dimensions

inch[mm]

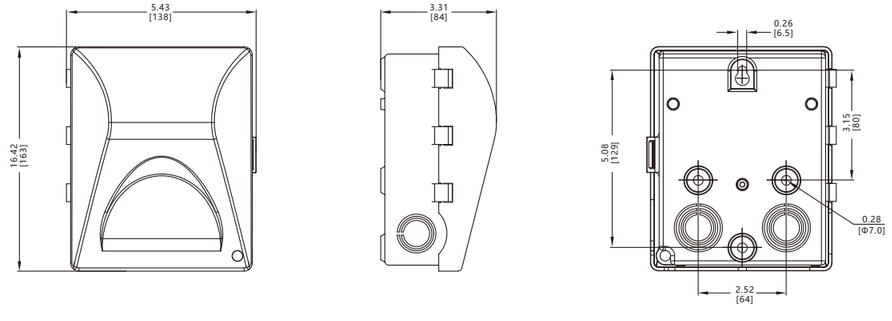
### PDS-60AN, PDS-60A, PDS-30A




### PCDS-60AN, PCDS-60A, PCDS-30A




### PCDS-60AN(Mini)



# HDS1 Air Conditioning Disconnects







## Product overview

The Most Widely Used Application for Air Conditioning Disconnects is for Residential and Light Commerical AirConditioning Units.An Air Conditioning Disconnects are also found in use with Heat Pumps, Swimming Pools,Spas, Whirlpools and Pump Houses and Required for Servicing Electric Water Heaters.

## Features & benefits

- ◇ Metallic and non-metallic options are NEMA 3R rated outdoor enclosures
- ◇ Easy-to-remove high-strength protective shield
- ◇ Easy-to-remove front cover
- ◇ Copper-rated line and load lugs
- ◇ Ample wiring space for mounting with a stud gun
- ◇ Padlock-able door provision for safety and reduction of tampering
- ◇ Factory-installed tamper resistant/weather-resistant ground fault receptacles

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 1429 |
| RoHS | RoHS | RoHS Compliant |

## Specification

Air Conditioning Disconnects

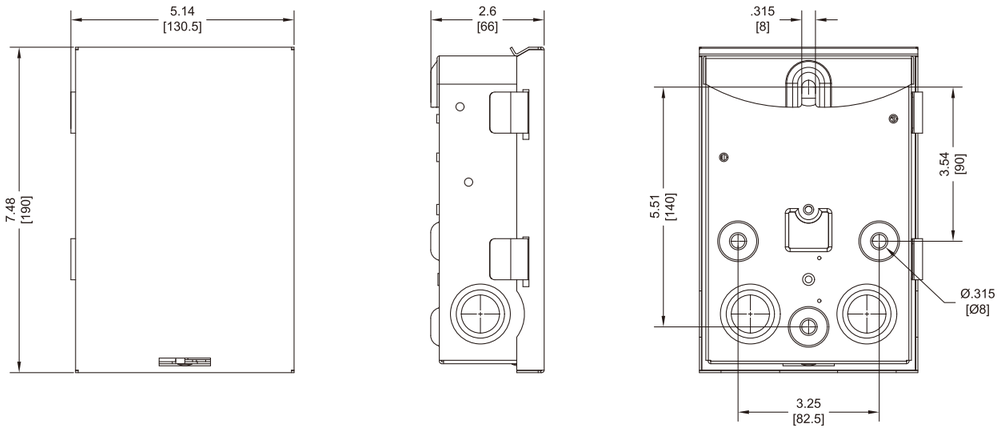
| Cat. No. | Description | Main Ampere Rating | HP Rating | | Wire Size(AWG) & Torque Cu/AL | |
|---|---|---|---|---|---|---|
| | | | 120V | 240V | LINE/LOAD | GROUND |
| HDS1-30A | Enclosed Fused ACD | 30AMP 240V<br>1 PHASE 2-Pole 3-WIRE | STD. 0.5<br>MAX. 2 | STD. 1.5<br>MAX. 3 | #14-10 35LB-IN<br>#8 40LB-IN<br>#6-4 45LB-IN<br>#2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |
| HDS1-60A | Enclosed Fused ACD | 60AMP 240V<br>1 PHASE 2-Pole 3-WIRE | STD. 1.5<br>MAX. 3 | STD. 3<br>MAX. 10 | #14-10 35LB-IN<br>#8 40LB-IN<br>#6-4 45LB-IN<br>#2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |
| HDS1-60AN | Enclosed Non-Fused ACD | 60AMP 240V<br>1 PHASE 2-Pole 3-WIRE | - | STD. 3<br>MAX. 10 | #14-10 35LB-IN<br>#8 40LB-IN<br>#6-4 45LB-IN<br>#2 50LB-IN | #14-10 25LB-IN<br>#8 40LB-IN |

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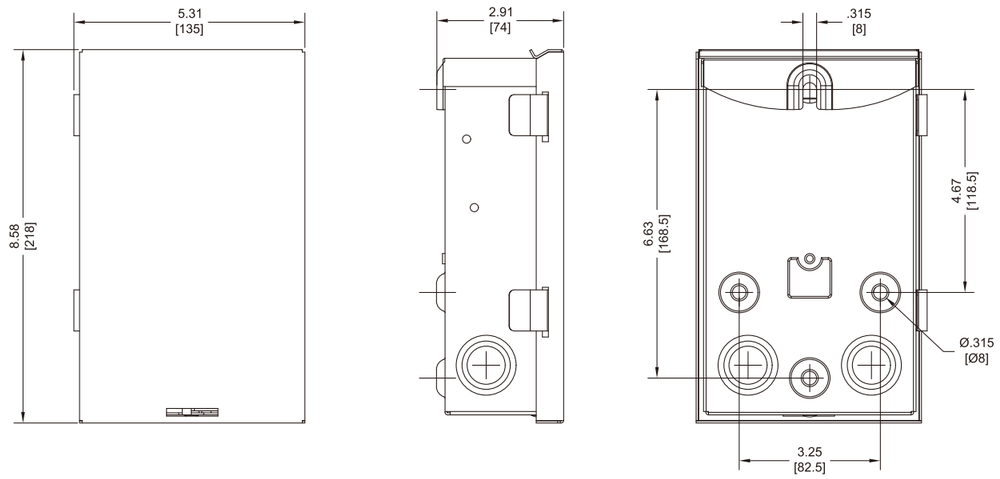
## Dimensions

inch[mm]

### HDS1-30A, HDS1-60AN

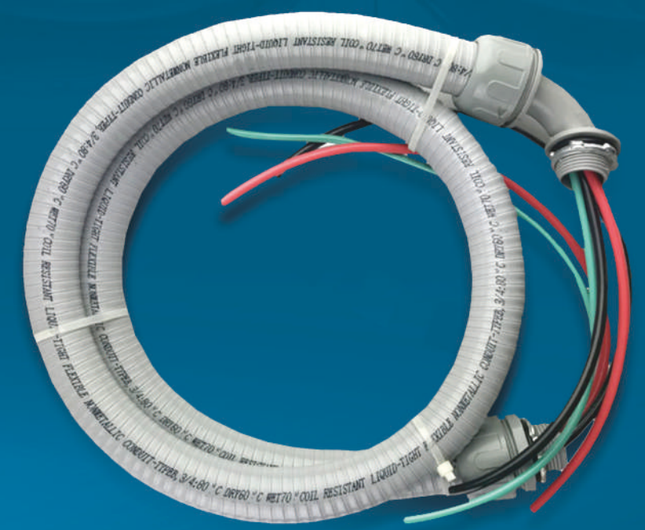



### HDS1-60A






# A/C Whips&Conduit



## Product overview

Ari Conditioner whips consists of UL listed flexible non-metallic conduit, one nonmetallic straight connector and onr nonmetallic 90° connector or one metallic straight connector and one metallic 90° connector.

## Application

◇Air Conditioner Wihps are typically used to connect the central

◇Air conditioner compressors to the disconnect box. Commonly used in air conditioner, pool and sauna

◇Installations

## Product certification

| | | |
|---|---|---|
| c UL us | UL | UL 60947-4-1 |
| RoHS | RoHS | RoHS Compliant |

## A/C Whips Non-Metallic Fittings

Non-metallic conduit with non-metallic fitting.



| NO. | DESCRIPTION | PACK |
|---|---|---|
| HC131 | 4'×1/2" 10AWG non-metallic whip UL | 6/Carton |
| HC133 | 4'×3/4" 8AWG non-metallic whip UL | 6/Carton |
| HC135 | 6'×1/2" 10AWG non-metallic whip UL | 6/Carton |
| HC137 | 6'×3/4" 10AWG non-metallic whip UL | 6/Carton |
| HC139 | 4'×3/4" 2×8AWG+1×10AWG non-metallic whip | 6/Carton |
| HC141 | 6×3/4" 2×8AWG+1×10AWG non-metallic whip | 6/Carton |

## A/C Whips Metallic Fittings

Non-metallic conduit with metallic fittings.



| NO. | DESCRIPTION | PACK |
|---|---|---|
| HC132 | 4'×1/2" 10AWG metallic whip UL | 6/Carton |
| HC134 | 4'×3/4" 8AWG metallic whip UL | 6/Carton |
| HC136 | 6'×1/2" 10AWG metallic whip UL | 6/Carton |
| HC138 | 6'×3/4" 10AWG metallic whip UL | 6/Carton |
| HC140 | 4'×3/4" 2×8AWG+1×10AWG metallic whip | 6/Carton |
| HC142 | 6'×3/4" 2×8AWG+1×10AWG metallic whip | 6/Carton |







## AC Whips Connector Reference Guide

Type: Straight, Screw-in

Material: Liquid Tight PVC

Specify Size:1/2", 3/4"





Type: 90° , Screw-in

Material: Liquid Tight PVC

Specify Size: 1/2", 3/4





Type: Straight, Screw-in

Material: Die Cast

Specify Size:1/2", 3/4"





Type: 90° , Screw-in

Material: Die Cast

Specify Size:1/2", 3/4"





# SPD Surge Protective Device






Hvacstar



## Product overview

- Protect HVAC equipment from electrical surges
- Protects from up to 25,000-amp single surges
- Type 1 Device can be mounted anywhere in the electrical system

## Recommended

- For Residential and commercialuse
- All single phase 150/300V AC outside condensing units and indoor air handlers

## Features

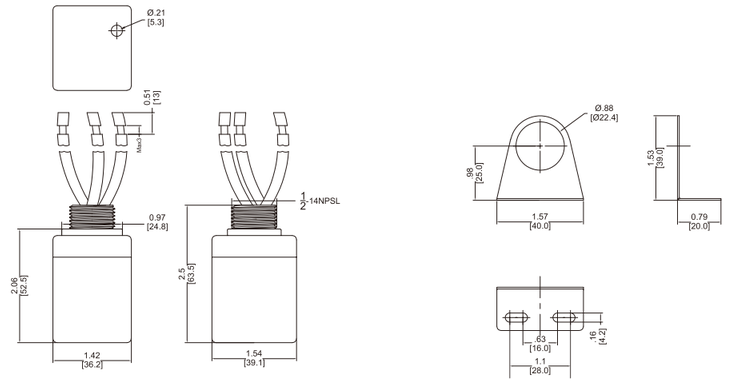
◇ TFMOV technology (thermaly fused metal oxide varistor)
◇ Limited lifetime warranty
◇ Green LED easily indicates protection
◇ Weather-Rated under NEMA4
◇ Fits all single-phase units
◇ UL1449 listed
◇ NEC's highest short circuit current rating(SCCR)@200kA
◇ RoHS compliant
◇ Temperature -40—85℃

## Specifications

| MODEL NUMBER | SPD-25 |
|---|---|
| SURGE CURRENT PER PHASE | 25KA |
| VOLT | 120/240V AC |
| PHASE | 1 |
| FREQUENCY | 50/60HZ |
| VPR (Voltage Protection Rating) | L-G:700V , L-L:1200V |
| MCOV | L-G:150V , L-L:300V |
| SCCR | 200KA |
| IN | 10KA |
| SPD | TYPE1 |

## Dimension

inch[mm]



# HDS2 Series Rainproof Power Outlet





## Product overview

HeCheng Electric delivers durable,dependable outdoor power where you need it. From building sites to barns, sheds and back yards, we help you power the great outdoors!

## Product feature

- ◇ Durable polyester powder coat finish resists chipping and fading
- ◇ Weather resistant and tamper resistant GFCI receptacles
- ◇ NEMA 3R weatherproof enclosure
- ◇ Compact design saves space
- ◇ Padlock provision helps prevent unauthorized access
- ◇ Cover desian allows cords to exit while maintaining NEMA 3R integrity
- ◇ Rolled edge door provides extra cord protection

## Specifications

| Model Number | Receptacles | Amps | Cabinet Size (in.) | Weight (lbs.) | |
|---|---|---|---|---|---|
| HDS2-001 | TT-30R | 125VAC 30A | 5×7 | 3.7 | Yes |
| HDS2-002 | 5-50R | 125VAC 50A | | 3.7 | |
| HDS2-003 | 14-50R | 125V/250VAV 50A | | 3.8 | |
| HDS2-004 | GFCI | 125VAC 20A GFCI TR/WR | | 3.7 | |
| HDS2-005 | GFCI*2 | 125VAC 20A GFCI TR/WR | | 4 | |
| HDS2-006 | TT-30R & GFCI | 125VAC 30A & 125VAC 20A GFCI TR/WR | | 4 | |
| HDS2-007 | GFCI | 125VAC 15A GFCI TR/WR | | 3.7 | |
| HDS2-008 | GFCI*2 | 125VAC 15A GFCI TR/WR | | 4 | |
| HDS2-009 | GFCI & Switch | 125VAC 15A GFCI TR/WR & 120VAC 15A Switch | | 3.9 | |
| HDS2-010 | GFCI & Switch | 125VAC 20A GFCI TR/WR & 120VAC 15A Switch | | 3.9 | |
| HDS2-011 | L14-30R | 125V/250VAC 30A | | 3.7 | |
| HDS2-012 | L5-30R | 125VAC 30A | | 3.7 | |
| HDS2-013 | L14-30R & Switch | 125V/250VAC 30A & 120V 15A Switch | | 3.9 | |
| HDS2-014 | L5-30R & Switch | 125VAC 30A & 120V 15A Switch | | 3.8 | |
| HDS2-015 | PDS & GFCI | 125VAC 15A GFCI TR/WR & PDS | | 3.9 | |
| HDS2-016 | PDS & GFCI | 125VAC 20A GFCI TR/WR & PDS | | 3.9 | |

## Use Copper Wire WIRE SIZE (AWG)

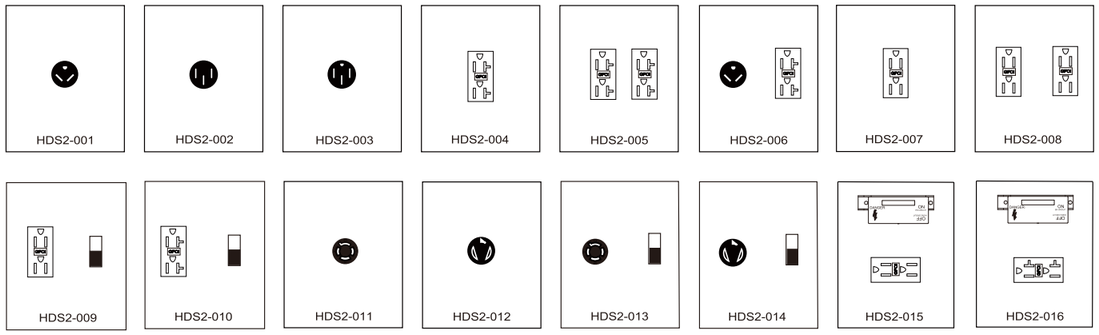
| Model Number / Terminal | HDS2-001/002 HDS2-003 | HDS2-004/005 HDS2-007/008 | HDS2-006 | | HDS2-009/010 | | HDS2-011/012 | HDS2-013/014 | | HDS2-015/016 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | TT-30R | GFCI | GFCI | Switch | | L14/L5-30R | Switch | PDS | GFCI |
| LINE | 10-4 | 14-12 | 10-4 | 14-12 | 14-12 | 14-12 | 14-8 | 14-8 | 14-12 | 14-2 | 14-12 |
| LOAD | 10-4 | 14-12 | 10-4 | 14-12 | 14-12 | 14-12 | 14-8 | 14-8 | 14-12 | 14-2 | 14-12 |
| Neutral/Ground | 10-4 | 14-12 | 10-4 | 14-12 | 14-12 | 14-12 | 14-8 | 14-8 | 14-12 | 14-2 | 14-12 |

Hvacstar

## Use Aluminum or Copper wire

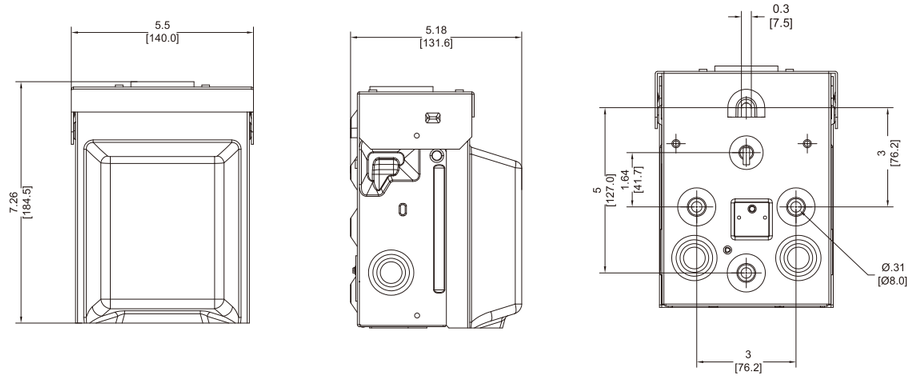
| | WIRE SIZE(AWG) | TORQUE(LB-IN) |
|---|---|---|
| Equipment Ground | 14-10 | 25 |
| Terminal | 8 | 40 |

## Layouts

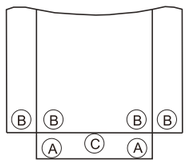



## Dimension

inch[mm]




Knockout




Knockout Key
A = 1/2", 3/4"
B = 1/2", 3/4", 1"
C = 1/2", 3/4", 1", 1-1/4"



# HCB1U-60 MCB



## Product overview

UL 489 Circuit Breaker provides excellent over-current, short-circuit protection and control in AC ON/OFF operating electrical systems, and can also be used for infrequent making and breaking.Widely used in HVAC equipment,Household appliances,industrial control cabinet and other industries.

## Product feature

- ◇ UL 489 listed full line of miniature circuit breakers
- ◇ Circuit breaker mountable on standard 35mm DIN rails with optional feet for front-and-bottom mounting
- ◇ The voltage is 120/240Vac, which meets the standard 10kA breaking current
- ◇ Suitable for protection of main circuit devices
- ◇ Contact position indication (red or green)
- ◇ Switch can be locked in the ON or OFF position
- ◇ Main circuit or branch protection

## Product certification

| | | |
|---|---|---|
| c UL us LISTED | **UL** | UL 489 |
| RoHS | **RoHS** | RoHS Compliant |

## Code explanation

| HC | B1U | -60 | C | 60 | /2 | MF |
|---|---|---|---|---|---|---|

Company Code

Model: UL 1 Series MCB

Frame size: 60

Tripping Curve: C: Type C Curves 8In±20%

Rated current(A):
5:5A 10: 10A 15: 15A 20: 20A 25: 25A 30: 30A
35: 35A 40: 40A 45: 45A 50: 50A 60: 60A

Pole: 1: 1P 2: 2P 3: 3P

Optional accessories, MF: mounting foot, SC:Safety cover, HS: Handle safety lock,HT: Hand Tie



Hvacstar

HCB1U-60 MCB

## Characteristic Parameters

| Pole | 1 Pole | 2 Poles | 3 Poles |
|---|---|---|---|
| Applicable standards | UL 489 | | |
| Rated Current(A) | 5~60 | | |
| Rated Voltage(V) | 120/240Vac | | 240Vac |
| Rated Frequency(Hz) | 50/60 | | |
| Tripping Characteristics | C(6.4In~9.6In) | | |
| Tripping Type | Thermal Magnetic | | |
| Protection Degree | IP30 | | |
| Electrical Life | 6000 | | |
| Mechanical Life | 10000 | | |
| Rated Insulation Voltage(Vac) | 1000 | | |
| Rated Impulse Withstand Voltage(kV) | 6 | | |
| Rated Short-curcuit Current(kA) | 10 | | |
| Operating Temperature Range | -25℃~+40℃ (Reference temperature 30℃) | | |
| Pollution Level | Class III | | |
| Elevation ft(m) | Not exceeding 6,561(2000) | | |
| Air Relative Humidity | 20℃: ≤90%, 40℃: ≤50% | | |
| Conducting Wire($mm^2$) | 14~4(2~25$mm^2$ 3.5N.m) | | |
| Installation | 35mm Din Rail Mounting, optional feet can be bottom mounted or surface mounded | | |

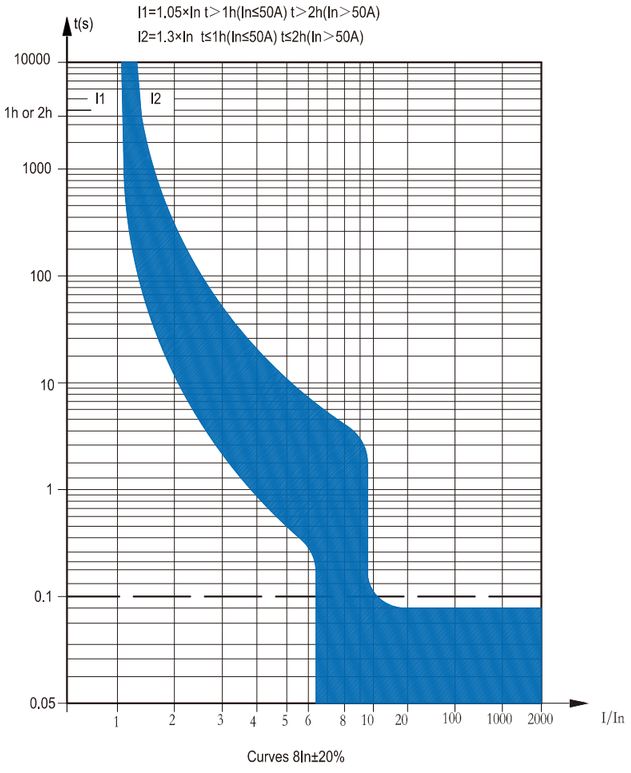
## UL Interrupting Rating

| Circuit Breaker Type | No.of poles | Ampere Rating | AC Volts | |
|---|---|---|---|---|
| | | | 120/240 | 240 |
| HCB1U-60 | 1 | 5~60 | 10kA | NA |
| | 2 | 5~60 | 10kA | NA |
| | 3 | 5~60 | NA | 10kA |

## Circuit breakers tripping curves

According to IEC/EN 60947-2 reference temperature 40 ℃（104°F）

AC TRIPPING CURVES




Curves 8In±20%



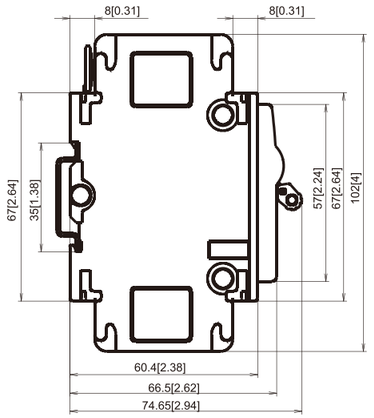


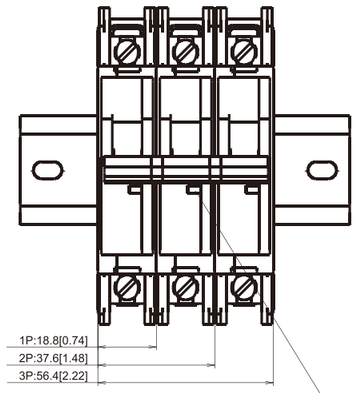


## Dimensions

mm[inch]

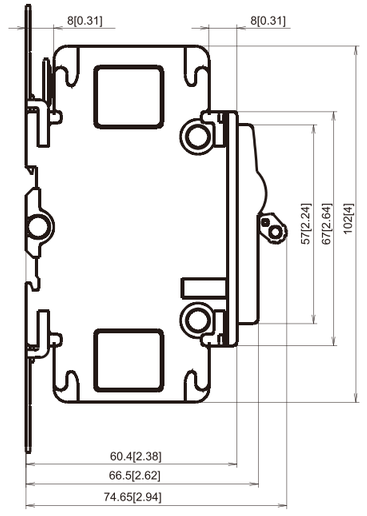
DIN rail mounting

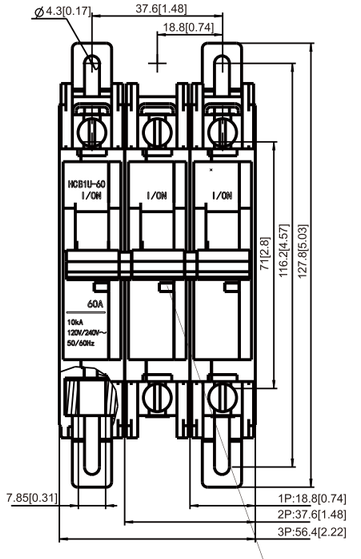



Visual switch window Open shows green, closed shows red.

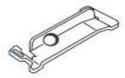
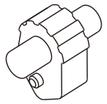
Mounting foot installation



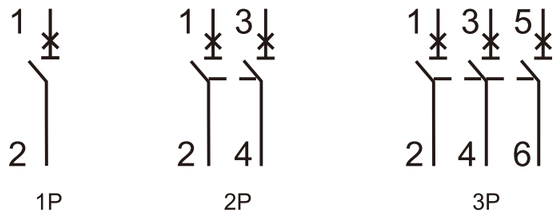

Visual switch window Open shows green, closed shows red.

## Optional accessories

| Accessories | Picture | Description | Dispostion |
|---|---|---|---|
| MF (Mounting feet) |  | Surface mount or base plate mount | 1Pole (1 Pair)<br>2Pole (2 Pair)<br>3Pole (2 Pair) |
| HS (Handle safety lock) |  | Handle locking attachment secures handle in on or offposition.<br>Each one is optional | 1 Set 1 Pcs |
| SC (Safety cover) |  | For circuit breaker inlet, one longsafety cover is required for each pole of circuit breaker.<br>Optional: one pair per pole<br>For example: HCB1U-60 / 3, optional 3 pairs | 1Pole (1 Pair)<br>2Pole (2 Pair)<br>3Pole (3 Pair) |
| HT (Hand Tie) |  | Tie two circuit breakers together. | 1 Set 1 Pcs |

## Diagram

HCB1U-60




## Precautions

A. The wiring must be reliable to prevent the circuit breaker from malfunctioning due to abnormal heating of the terminal or causing the terminal to burn.

B. The nut wiring specification of this product is 14-4AWG, and the tightening torque during installation is 3.5 N.m

C. For multi-pole products, the handle should be operated in the middle position, and the time should be less than 1s when the product is normally opened and closed.

D. 1-pole product weight: 0.161Kg, 2-pole product weight: 0.314Kg; 3-pole product weight: 0.467Kg.

E. Warranty policy: Product sales date starts to calculate, due to quality problems caused by product failure, non-human damage, etc., send back free replacement, warranty period of 24 months.
Artificial damage, out of warranty, etc



# Class RK5 Fuses



# Product overview

Dual-element fuses can be applied in circuits subject to temporary motor overloads and surge currents to provide both high-performance short-circuit and overload protection.The overload element provides protection against low-level overcurrent ofoverloads and will hold an overload which is five times greater than the amprating of the fuse for a minimum of ten seconds.

## Features

Dual-element fuses can be applied in circuits subject to temporary motor overloads and surge currents to provide both high performance short-circuit and overloadprotection.The overload element provides protection against low-level overcurrent of overloads and will hold an overload which is five times greater than the amp rating of the fuse for a minimum of ten seconds.

## Product certification

| | | |
|---|---|---|
| UL | **UL** | UL 248 |
| RoHS | **RoHS** | RoHS Compliant |
| Reach | **REACH** | REACH Compliant |

## Fuse Ratings

Time-Delay – 10 seconds (minimum) at 500% rated current (8 seconds for fuses 30A or less)

HCNR (10-30A)

**HCNR010 HCNR012 HCNR015 HCNR020 HCNR025 HCNR030**

HCNR (35-60A)

**HCNR035 HCNR040 HCNR045 HCNR050 HCNR060**

## Current-Limiting Effects of HCNR Fuses

| SHORT CIRCUIT CURRENT* | APPARENT RMS SYMMETRICAL CURRENT FOR VARIOUS FUSE RATINGS | |
|---|---|---|
| | 30A | 60A |
| 5000 | 1400 | 2100 |
| 10000 | 1550 | 2500 |
| 15000 | 2000 | 3150 |
| 20000 | 2250 | 3400 |
| 25000 | 2400 | 3750 |
| 30000 | 2550 | 4100 |
| 35000 | 2650 | 4300 |
| 40000 | 2800 | 4400 |
| 50000 | 3000 | 5000 |
| 60000 | 3200 | 5250 |
| 80000 | 3400 | 5750 |
| 100000 | 3850 | 6000 |
| 150000 | 4100 | 7000 |
| 200000 | 4300 | 7500 |

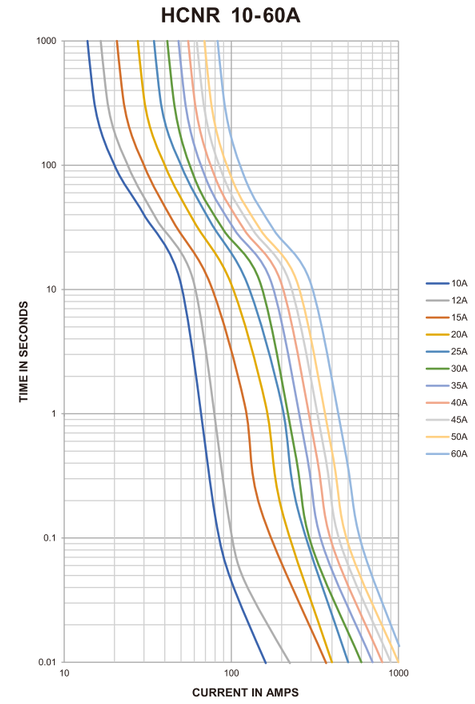
*Prospective RMS Symmetrical Amperes Short-Circuit Current
Note: Data Derived from Peak Let-Thru Curves





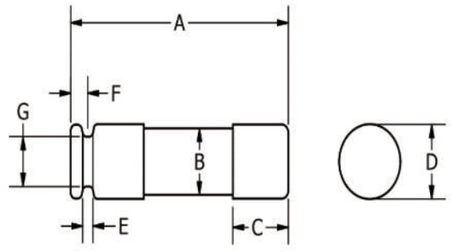


## Time-Current Curve




## Dimension

mm




| Rated Current | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| | ±1mm | ±1mm | ±0.5mm | ±0.5mm | ±0.5mm | ±0.5mm | ±0.5mm |
| 10-30A | 50.8 | 12.7 | 12.7 | 14.3 | 1.8 | 4.0 | 9.5 |
| 35-60A | 76.2 | 19.2 | 15.9 | 20.7 | 2.4 | 4.7 | 15.9 |